Fast Simulation-Based Bayesian Estimation of Heterogeneous and Representative Agent Models

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This paper proposes a simulation-based maximum likelihood or Bayesian algorithm for the estimation of macroeconomic models. This approach is able to derive posteriors or the maximum likelihood estimate even when the likelihood function is not tractable. Because the likelihood is not needed for Bayesian estimation, filtering is also not needed. The approach shares similarity to (Kristensen and Shin, 2012), but it replaces the kernel density estimator with the option of a few different machine learning density estimators that can deal with high dimensions while being numerically stable. I demonstrate the validity of the approach by estimating a 10 parameter HANK model solved via Reiter's method that generates 812 covariates per time step, where 810 are latent variables, showing this can handle a large latent space without state space reduction. I also estimate the algorithm with an 11-parameter model solved via value function iteration, which cannot be estimated with Metropolis-Hastings or even conventional maximum likelihood estimators without measurement errors. In addition, I show the posteriors estimated on Smets-Wouters 2007 are higher quality and faster using simulation-based inference compared to Metropolis-Hastings. This approach helps address the computational expense of Metropolis-Hastings and allows solution methods which don't yield a tractable likelihood to be better estimated.

JEL Codes: E27, E37, E47, C45, C11, C15
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I. Introduction

In the field of economics, the estimation of many structural models remains a challenge due to the difficulty in estimating likelihoods in many situations. This has resulted in the reliance on simulation-based techniques, despite the advantages of maximum likelihood estimation (MLE) if it is possible to use. The advantages of maximum likelihoood include efficiency, the fixed nature of the objective function, and compatibility with Bayesian methods. The seminal work of McFadden (1989) on the method of simulated moments marked the starting point for simulation-based estimation in structural models, with subsequent works by Smith Jr (1993), Kaji, Manresa and Pouliot (2020), and Gourieroux, Monfort and Renault (1993) aimed at improving its performance. Despite efforts to improve simulation-based approaches, they continue to lack these advantages. There has been some work at unifying these two approaches, but the work has failed to scale to state of the art models. For example, Kristensen and Shin (2012) proposed a likelihood-based approach using a kernel density estimator, but it was limited to models with only a few unimodal parameters and lacked a Bayesian extension. Likewise Kaji, Manresa and Pouliot (2020) proposed using a Generative Adversarial Network (GAN) (Goodfellow et al., 2014), however this approach lacks a Bayesian extension and imposes a Markov/cross-sectional assumption on the data generating process.

This paper presents a simulation-based likelihood method for estimating structural models that combines the advantages of both approaches, solving the above problems. Following Kristensen and Shin (2012), the method begins by simulating from the prior, $P(\theta)$ and then drawing simulated data, X', from the model conditional on the priorto generate samples from the joint distribution, $X', \theta \sim P(X', \theta) = P(X'|\theta)P(\theta)$. A machine learning density estimator, such as a conditional mixture of Gaussians (Bishop, 1994), a normalizing flow (Rezende and Mohamed, 2015), or GAN (Goodfellow et al., 2014), is then used to estimate the conditional density $P(\theta|X')$ from the joint samples $(X', \theta, \sim P(X', \theta)$. The density estimator is then conditioned on the observed X data, giving an estimate of the posterior: $P(\theta|X' = X)$. These machine learning techniques can handle high-dimensional density estimation, allowing for the macroeconomic data to be treated as a single data point, which permits the imposition of the state space assumption that all past data has at

least a small but non-zero effect on current data. This is in contrast to current simulation-based approaches that impose a Markov/cross-sectional assumption on the data as each time step is only effected by a finite number of lags.

The proposed method is implemented in a Bayesian framework, but the posteriors can be easily converted to MLE estimates by using uniform priors and taking the mode of the posterior. As a Bayesian algorithm, the approach has several attractive properties compared to traditional Metropolis-Hastings Markov Chain Monte Carlo (MH-MCMC) estimation, which is computationally inefficient, especially for slow state-of-the-art models (Herbst and Schorfheide, 2015). Furthermore, the development of other techniques, such as variational inference, have raised concerns about the adequate convergence of MH-MCMC on complex models (Mohammad-Djafari and Ayasso, 2009). Results suggest that even after 10 million samples from the Smets and Wouters (2007) model, MH-MCMC has not fully converged, as the posterior of many parameters does not include the ground truth parameter in the support of the distribution. In contrast, the proposed simulation-based algorithm, relying on density estimation rather than sampling from the posterior, has better-behaved posteriors even after 150 thousand samples from solving the model, a reduction of almost two orders of magnitude in the number of samples required, although each sample takes more time to estimate. My algorithm performs the simulations by calling Matlab from Python, a slower process than the MH-MCMC which is performed using optimized Dynare Matlab code. Despite 66x less simulations, the posteriors obtained using SBI are often near the calibrated true parameter values, whereas those obtained using MH-MCMC are almost never close to the true values.

In both the MLE and Bayesian setting, the use of SBI is particularly beneficial for models with a large state space, such as the HANK model, as it avoids the computational intractability of direct Kalman filtering or particle filtering. The end-to-end training of a dimensionality reduction neural network in conjunction with the density estimator allows for the recovery of a lower-dimensional sufficient statistic that conveys all the information required for Bayesian estimation. This sufficient statistic should exist because the simulated macroeconomic data is generated from a lower dimensional parameter space. In comparison, PCA dimensional reduction, as advocated by (Ahn et al.,

2018), loses information as it is not trained end-to-end with the Bayesian objective. Additionally, the approach can perform Bayesian and MLE estimation without the use of measurement error in the case of projection solution techniques, and may be able to do the same with value function solution methods.

The general approach around simulation-based inference has gained popularity in many fields. The approach is often used for Bayesian inference in machine learning (Durkan, Murray and Papamakarios, 2020), (Greenberg, Nonnenmacher and Macke, 2019). The approach has also been used in neuroscience (Boelts et al., 2021) and ecology (DiNapoli et al., 2021). Furthermore, the technique has also become one of the leading estimation methods in many fields of physics (Brehmer, 2021), (Cranmer, Brehmer and Louppe, 2020). Economics could benefit from adopting such a widely proven technique.

II. Literature Review

The aim of this literature review is to provide a comprehensive overview of three topics in economics. The first topic is simulation-based methods in economics, including their history, development, and applications. The second topic focuses on the literature surrounding the solution and estimation of dynamic models with kinks and discontinuities. Finally, the review will delve into the solution and estimation of heterogeneous agent models, including their relevance and current state of the literature. I will cover the concepts of MH-MCMC and machine learning and their applications in the section following this literature review.

II.A. Simulation-Based Models in Economics

Simulation-based estimation has been a prominent approach in economics for many years. Originating from the method of simulated moments (MSM) (McFadden, 1989), simulation-based likelihood and inference techniques have been developed (Pakes and Pollard, 1989), (Duffie and Singleton, 1990) and improved to address the difficulty of improving efficiency. Maximum simulated likelihood (Lerman and Manski, 1981) (Hajivassiliou et al., 1997), (Haan and Uhlendorff, 2006) and

other techniques have been used to improve efficiency, but still require the use of measurement error. There has also been work on using simulation-based methods to solve dynamic models, particularly in industrial organization (Keane and Wolpin, 1994), as well as on efficient simulation techniques using indirect inference (Smith Jr, 1993), (Gourieroux, Monfort and Renault, 1993) and efficient method of moments (Gallant and Tauchen, 1996).

Recent research in simulation-based techniques has aimed to improve robustness to misspecification. Efforts have been made to develop efficient method of moment estimators using a spectrum of moments to guarantee that data is in the support of distributions, but these methods impose a Markovian data generating process assumption. Another approach is approximate Bayesian inference (Rubin, 1984), but this is only approximately Bayesian in a well-specified model.

Particle filtering has also been considered a simulation-based technique for deriving a likelihood function (Fernández-Villaverde and Rubio-Ramírez, 2007), but it only works with dynamic models that yield complementary state space representations and requires a likelihood function at each point in time. While some work has been done on simulated Bayesian inference (Flury and Shephard, 2011) (Herbst and Schorfheide, 2014) and GANs for machine learning Kaji, Manresa and Pouliot (2020), there is still an open question in the simulation-based literature in economics for the development of efficient simulation-based inference tools that assume the data follows a state space or long memory data generating process, rather than a Markov assumption. There are methods that allow for efficient estimation via simulation (Kaji, Manresa and Pouliot, 2020), but don't impose the state space assumption, and others that allow for the use of particle filter likelihoods (Herbst and Schorfheide, 2014), but will not work with projection or value function iteration solution techniques without point by point likelihoods. This paper aims to fill this gap by proposing a method that can efficiently estimate structural models solved in any way via simulation while also maintaining fidelity to the state space data generating process assumption.

II.B. Solving Dynamic Models an Example of an Area of Need: Kinked Models

Solution methods for dynamic macroeconomic models present many challenge for estimation. Three widely utilized methods for solving these models are perturbation, projection, and value function iteration, each with its own strengths and limitations (Fernández-Villaverde, Rubio-Ramírez and Schorfheide, 2016), (Judd et al., 2017). Perturbation is well-suited for models where the policy function appears as a linear or low-degree polynomial function and yields likelihood functions that allow for Bayesian and MLE estimation using conventional techniques. However, for highly nonlinear models with large shocks or non-analytic functions, projection and value function iteration are preferred (Terry, 2017). Unfortunately, for a significant number of models including those solved via value function iteration (Hennessy and Whited, 2007), heterogeneous agent models with intractable likelihoods (Kukackaa and Barunika, n.d.), and most projection models, the likelihood is intractable and practitioners must rely on less efficient point estimate methods that are subject to limitations compared to Bayesian and MLE estimation. The estimation of non-perturbation models without measurement error requires simulation-based inference approaches, typically resulting in non-Bayesian and non-full information estimations, requiring the use of methods such as MSM.

As an illustration of problems faced by a perturbation approach, below is an example of a s-S model policy function in Caballero and Engel (2007):

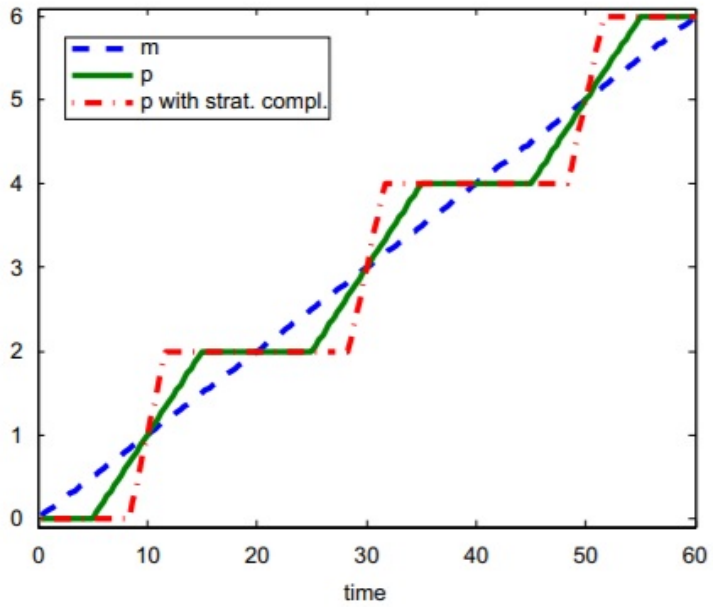

Fig. 2. p and m in an extension of Caplin–Spulber.

FIGURE I: s-S MODEL

The s-S model framework postulates a fixed cost associated with changing a state variable, leading to the emergence of kinks in the policy function. This assumption is often necessary for modeling inventory and capital. For instance, in the above picture of menu prices, the s-S cost with respect to a linearly increasing money supply leads to a spectrum of firm prices ranging from the s-S trigger point to zero (Caplin and Spulber, 1987). Caballero and Engel (2007) assume there is a spectrum of firms prices that only covers a portion of the range from the s-S trigger to zero. As the money supply increases, there are periods where no firms adjust due to the fixed cost and other periods where an above-average number of firms adjust to keep up with the price level. This results in the policy function being non-differentiable and non-analytic, rendering a perturbation approach, which relies on a Taylor series approximation, ineffective. Therefore, value function iteration or similar methods must be used to solve the model, but these methods present estimation difficulties as previously discussed.

The use of kinked value functions can be found in various areas of economics, including heterogeneous agent models for investment in capital Khan and Thomas (2008), purchasing behavior for used durable goods House and Leahy (2004), and structural models in finance (He, Whited and Guo, 2021), (Taylor, 2013), (Terry, Whited and Zakolyukina, 2022). Despite the demand for efficient and potentially Bayesian methods for estimation of these kinked and other non-analytic models, the literature lacks the ability to estimate these models without measurement error, providing an important area for this estimation technique

II.C. Heterogeneous Agent Models

The literature surrounding full information estimation of heterogeneous agent models is vast and growing, with this section will focus on improving speed of estimation methods. Researcher emphasize computational speed, with many papers proposing faster solution methods (Auclert et al., 2021*a*), (Winberry, 2018), (Khan and Thomas, 2008). However, at least on the heterogenous agents side, there is demand for faster estimation approaches. Despite the computational challenges posed by the intractability of the Kalman filter in large latent spaces, economists continue to work on full information estimation techniques. For example, Ahn et al. (2018) proposes to use a PCA based approach to reduce the dimensionality of the latent space.

Other innovations include Liu and Plagborg-Møller (2023) which uses full information microdata to complement macro variable estimation to perform true full-information Bayesian estimation. Another, by Parra-Alvarez, Posch and Wang (2020), estimates an Aiyagari model (Aiyagari, 1994), (Huggett, 1993), but highlights the issue of poor parameter identification and suggests calibration as a solution. This paper finds similar issues estimating the Winberry (2018) model in the spirit of Liu and Plagborg-Møller (2023) on empirical data. For this reason among others, while there is a large literature on heterogeneous agent model (Auclert et al., 2021*b*), (Ottonello and Winberry, 2020), (Caglio, Darst and Kalemli-Özcan, 2021), most practitioners resort to calibration. There is a need among both heterogeneous agent and representative agent model practitioners to reduce the barriers to maximum likelihood estimation and Bayesian estimation, and the SNPE and related algorithms may help address these barriers.

III. Background on Simulation-Based Bayesian Inference (SBI) estimator and Metropolis-Hastings

In recent years, Bayesian inference has gained increased prominence as a method for estimating models in economics. This section sheds light on two prominent techniques in this field, namely the SBI estimator and MH-MCMC. To provide a comprehensive understanding, a brief overview of Bayesian estimation is first presented. The SBI estimator, which relies on simulating data from the model being estimated and calculating the posterior using machine learning density estimators, is then discussed. Three approaches can be considered: a mixture density model (Bishop, 1994), a normalizing flow (Mohammad-Djafari and Ayasso, 2009), or a GAN approach (Goodfellow et al., 2014). The SBI estimator is an efficient method, requiring fewer simulations than methods such as MH-MCMC.

Despite its success in various applications, the last section will discuss the limitations and pathological cases for the SBI estimator. While these methods have been widely used, it is essential to understand their limitations and make informed decisions when choosing the most appropriate approach for a particular problem, considering the assumptions and limitations of each method.

III.A. Bayesian Basics

In this subsection, I will discuss the general paradigm of Bayesian estimation, then I will discuss MH-MCMC.

Bayesian estimation attempts to find the posterior given the likelihood and prior of a model using Bayes' rule:

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$$

Here $P(\theta|X)$ is the posterior, $P(X|\theta)$ is the likelihood, and $P(\theta)$ is the prior. All approaches that perform Bayesian inference, ranging from MH-MCMC, variational inference, to simulation-based inference concern techniques for calculating $P(\theta|X)$ without calculating the partition function, $P(X)$.

MH-MCMC is a commonly employed technique for Bayesian inference in economics, as described in (Herbst and Schorfheide, 2015). The way MH-MCMC gets around the intractable partion function is to evaluate the likelihood ratios between different parameter values, allowing for the determination of the relative likelihood of different points in the parameter space, without calculating $P(X)$. This information is then utilized to direct a random walk, with points being visited in proportion to the relative likelihoods, effectively sampling from the posterior distribution $P(\theta|X)$. If the prior and likelihood for a point are known, the relative likelihood of different points in the parameter space can be determined. This information is used to direct a random walk with the aim of visiting points in proportion to their relative likelihoods, effectively sampling from the posterior distribution $P(\theta|X)$. The predominant method of Bayesian inference in economics is the use of MH-MCMC (Herbst and Schorfheide, 2015).

III.B. Sequential Neural Posterior Estimation (SNPE)

The SNPE algorithm which is an extension of the Kristensen and Shin (2012), forms the foundation of this paper. This approach will be discussed in detail. In later subsections, I will discuss the choice of density estimator. Kristensen and Shin (2012) use a kernel density estimator, but using more powerful machine learning density estimators will significantly enhance the results.

This paragraph explains the details of the SNPE algorithm. The algorithm commences by drawing a sample, θ, from the prior distribution, $P(\theta)$. Subsequently, simulated data, X', is generated from the model likelihood, $P(X'|\theta)$. The concatenation of these two simulations results in samples from the joint distribution, $X', \theta \sim P(X', \theta) = P(X'|\theta)P(\theta)$. These samples can then be used to estimate the posterior distribution, $P(\theta|X')$, through the utilization of either a normalizing flow density estimator (Mohammad-Djafari and Ayasso, 2009) or a conditional mixture of Gaussians (Bishop, 1994). The former will be described in the appendix, while the latter will be discussed in the following section.

The aim of reducing variance in the estimation process can be achieved by performing multi-round inference in the SNPE algorithm. In this approach, θ is sampled from the best approximation of the posterior, leading to a more accurate density estimator. However, this process requires

importance sampling to correct for the shift in distribution as θ is no longer being drawn from the prior. The first round entails sampling from the prior, while later rounds involve sampling from the updated estimate of the posterior and applying importance sampling to account for the distribution shift. This can be mathematically expressed as $p'(\theta|X') = p(\theta|X')\frac{p'(\theta)}{p(\theta)}$, where $p'(\theta)$ is the proposed prior, $p(\theta)$ is the true prior, and $p'(\theta|X')$ is the estimated posterior obtained by sampling from the proposal distribution. The use of a flexible machine learning density estimator, such as a normalizing flow or a conditional mixture of Gaussians, allows for the unbiased estimation of the posterior through the estimation of $f_\phi(\theta|X')$ in place of $p(\theta|X')$ in the above equation. Papamakarios and Murray (2016) and Greenberg, Nonnenmacher and Macke (2019) have demonstrated that the use of universal approximators, such as normalizing flows (Huang et al., 2018) and conditional mixture of Gaussians (Bishop, 1994), as density estimators in multi-round SNPE allows for the approach to converge towards the true Bayesian posterior in the computational limit under mild conditions. Additionally, this paper uses the Sequential Neural Variational Inference (SNVI) in one example (Glöckler, Deistler and Macke, n.d.). SNVI employs a GAN to estimate the likelihood and projects it onto a normalizing flow. However, this approach is an extension to the SNPE algorithm and is discussed in the appendix.

Algorithm 1: SNPE Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, flow $f_\phi(x|\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim p^{(i-1)}$ for $n = 1...S$ with Monte Carlo;

  Simulate $X'^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{X'^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(X', \theta)$ *not converged* **do**

    Sample $\{X'^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Train $f_\phi(\theta|X')\dfrac{p^{(i-1)}}{p(\theta)}$ on $\{X'^{(i)}, \theta^{(i)}\}_i^B$;

  end

  Update posterior $p^{(i)} \propto f_\phi(\theta|X' = X)$;

end

III.C. Density Estimation

The SNPE and SNVI approaches proposed in this article utilize three distinct density estimators: Normalizing Flows, Conditional Mixture of Gaussians, and GANs. These methods are capable of handling high dimensional estimation in X' and θ. The structural models are used to generate samples from X' and θ, then with the exception of GANs, which uses the slightly different SNVI algorithm, each density estimator serves as the $f_\phi(\theta|X')$ function in the above algorithm.

One of the primary benefits of the proposed estimation algorithm is the improved handling of high dimensional estimation of θ. This allows for Bayesian and MLE estimation of models with up to 36 parameters, such as the Smets and Wouters (2007) model. This is a significant improvement over previous works, such as Kristensen and Shin (2012), which are limited in their ability to handle high dimensional estimation. Additionally, this algorithm treats the entire data, X, as coming from a state space, meaning that data from the distant past can still have a non-negligible impact on the current time step. This approach is distinct from simulation-based estimation techniques, which

typically treat data as a Markov process and only consider a limited number of lags for the current time step. This, in addition to a Bayesian extension, are the improvements over Kaji, Manresa and Pouliot (2020).

Each of the three density estimator approaches has its own strengths and weaknesses. Normalizing flows are highly accurate in fitting exotic distributions, including posteriors with discontinuous support or high but non-linear mutual information. On the other hand, mixture of Gaussians is the simplest model to implement and can handle most Bayesian posteriors with smooth and multimodal distributions. The SNVI/GAN approach is designed specifically for structural models that fail to converge regularly within the parameter bounds. Unlike the other two methods, which fit a density estimator directly on the data, the GAN approach aims to mimic the real data, thus placing zero probability mass in areas in the parameter set but lack training data, due to for example non-convergence of the solution method. The other methods will extrapolate pdf values in areas without training data (but in the training data range), potentially behaving strangely in these areas. This results in a slower and less accurate method, but with the benefit of disciplining distributions in areas with no training data.

III.D. Conditional Mixture of Gaussians

The conditional mixture of Gaussians approach is a powerful tool for density estimation in high dimensional spaces and, of the three methods, is the easiest to explain. As outlined in Bishop (1994), a mixture of Gaussians is a linear combination of N Gaussian distributions, each with its own mean, covariance, and weight. The probability density function (pdf) of a point under this mixture can be calculated using the equation:

$$P(\theta) = \sum_i \pi_i * Q_i(\theta)$$

Where $Q_i(\theta)$ is a normal distribution with mean and variance (covariance) specific to each Gaussian component. This represents the unconditional density estimator, however, the algorithm requires a conditional density estimator. This can be achieved by using an optimizable function,

such as a neural network, that takes the X' data as input and returns the parameters for the mixture of Gaussians. The network would return the weights $\pi_i(X')$, means $\mu(X')$, and covariances $\sigma(X')$ for each Gaussian component, resulting in the following conditional density estimator:

$$P(\theta|X') = \sum_i \pi_i(X') * Q_i(\theta|\mu(X'), \sigma(X'))$$

Finally here is an estimation of conditional density estimators. The parameters of the neural network can be optimized via maximum likelihood estimation (MLE), maximizing $P(\theta|X')$ and improving the accuracy of the density estimator. The approach maximizes $P(\theta|X')$ given the simulated conditioning data X'. The X' is inserted into the model, a feedforward neural network will learn a mapping from X' to the parameters of the mixture of Guassinas (ie $\sigma(X')$, $\mu(X')$ etc.). The parameters of the feedforward neural network is then optimized so that the mixture of Guassians produces the highest likelihood for the θ variables.

III.E. Caveats and Pathological Examples Using SNPE

There are certain limitations associated with SNPE that must be considered when applying it to a specific problem. One of the limitations of SNPE is its inability to handle the discontinuities that are present in some models with kinks, such as s-S models. This leads to a lack of smoothness in the posterior of the kinked models, and the assumption that there is a normalizing flow parameterization that can generate the posterior is violated. In practice, a flow still converges, but it learns a continuous function that approximates the discontinuity in the posterior. This approximation can still be near the true likelihood, but if estimation with a kinked model is a problem, one can also add measurement error to smooth the problem out.

Another limitation of SNPE is its difficulty in handling stochastic singularities, which also result in estimation issues. When this occurs, the dynamic model learns a sub-manifold in the entire space, and the normalizing flow or mixture of Gaussians has to have positive support on the entire space of the target distribution. This also violates the assumption that the machine learning density estimators can generate the posterior. In cases where there are stochastic singularities in a model

estimated on real data, the model will fail to match the data. However, one can add more shocks, add measurement error, or make a change in modeling to avoid a stochastic singularity (Caballero and Engel, 2007) and allow full support over the range of the model output.

IV. Results

This section will evaluate the performance of SNPE by examining the proper behavior of the posterior and inspecting its accuracy. To this end, seven models are estimated using the SNPE algorithm, each showcasing a unique aspect and characteristic of the algorithm. The first model is the RBC model that is estimated on both simulated and real data. The second model is a toy corporate finance model that is solved via value function iteration, and three different SBI approaches are utilized for its estimation. The third model is the Lucas asset pricing model solved through projection. The fourth model is the Smets-Wouters model, which is estimated through both the MH-MCMC and SNPE algorithms. The fifth model is the HANK model solved through Reiter's method and time iteration. The sixth model is a bequests model solved via value function iteration. The seventh model improves on full information estimation of a Winberry (2018) model speeding up the approach taken by Liu and Plagborg-Møller (2023). The details of the construction and implementation of these models can be found in the appendix.

In the empirical analysis, the accuracy of the posterior is inspected by comparing it with the calibrated parameters producing the simulated data. In theory, through the Bernstein-von Mises theorem, the posterior should converge to the true values, although perhaps the structural models may not satisfy the assumptions of the theorem. Although the correct posterior should converge to the true parameter value as the number of data samples increases, finite sample sizes may not always result in a perfect match. Hence, it is important to assess the posterior mode and ensure that it generally matches the true value and that there is no vanishing posterior mass at the true values. This check provides useful information to determine the performance of SNPE in practice and how it compares with other estimation techniques, such as the MH-MCMC algorithm. While in many cases the method is tested using simulated data, in order to resemble real world data, 200

time steps are sampled from the simulated model, to maintain a rough correspondence to the length of real world macroeconomic data. This small number of time steps implies that some parameters will be inadequately identified especially in the models with a large number of parameters. For the sake of illustrating the estimation benefits, all priors will be uniform over the range specified in the graph, making the estimation problem as difficult for the algorithm as possible.

IV.A. RBC Model

In the present article, the results of estimating the RBC model using the SNPE algorithm are presented and discussed. The RBC model is a standard model in macroeconomics and it has four parameters, α, β, δ, and ρ, with a fixed CRRA elasticity of 2. The first 100 iterations are dropped to attain a steady-state behavior. The results of the posterior are displayed below, where the red bar represents the true parameter values and the blue line represents the posterior as estimated by the simulation-based inference approach:

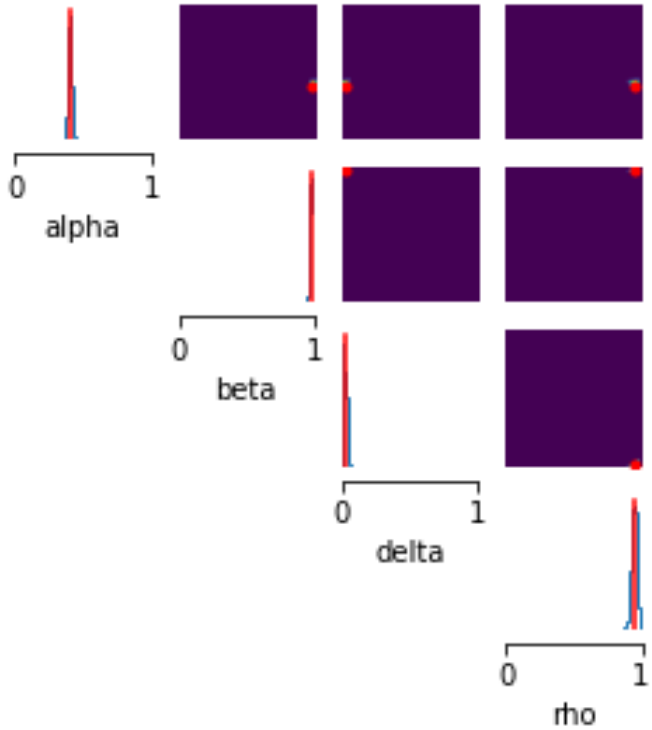

Figure II: Simulated Data RBC

The diagonal density plots in the upper triangular portion of the graph display two-way densities, with the color indicating the probability mass. The approximate ground truth is obtained by running the MH-MCMC for 2 million iterations as show below:

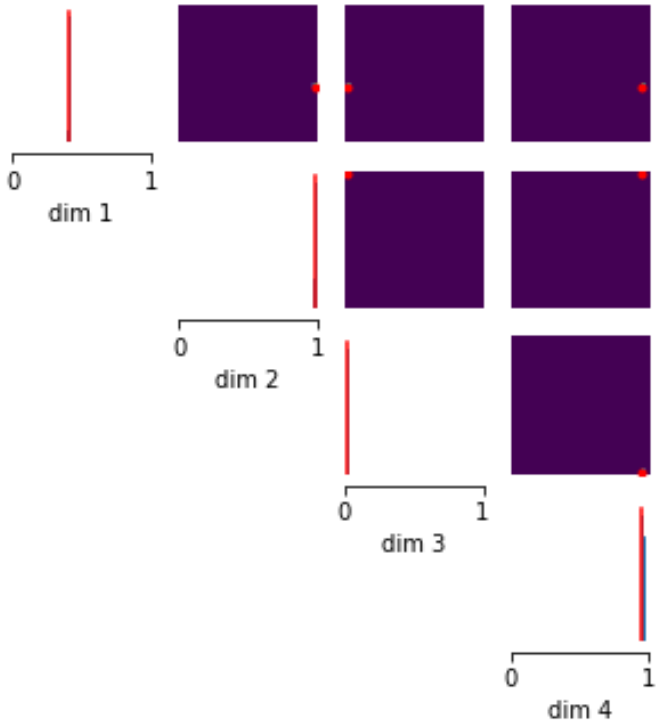

Figure III: Simulated Data RBC

Below the same RBC model is also estimated on real data, where there is no available ground truth. The parameters derived from the real data are not necessarily realistic, which is expected as the RBC model is a simple model. The method is uncertain about the ground truth, as indicated by the spread of the posterior. The model also predicts some parameters that are not in line with theory, such as the discount rate, β, and the parameter on the productivity process, ρ. The depreciation rate, δ, is the only parameter that is relatively reasonable. Below is the posterior estimated on real data:

In conclusion, the SNPE algorithm is applied to estimate the RBC model and the results are displayed and discussed. The results indicate that the SNPE algorithm can estimate the RBC model to a good degree of accuracy. The model's limitations in real-world applications are also highlighted, as it fails to estimate some parameters that are not in line with theoretical predictions.

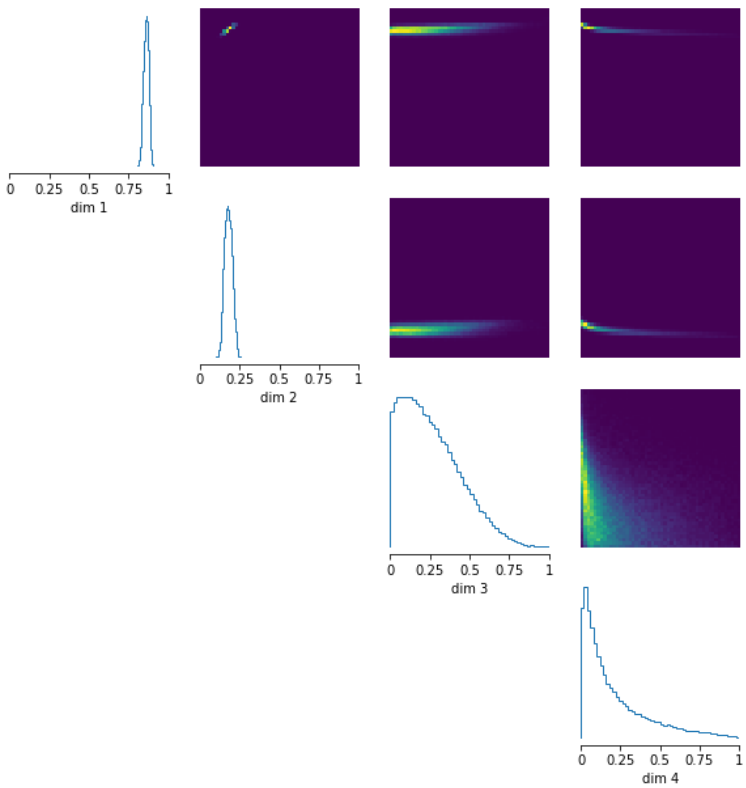

Figure IV: Real Data RBC

This highlights the need for more complex models to better reflect real-world economic scenarios, which will be illustrated in the next models.

IV.B. Cash Flow Model

In this study, we aim to estimate the partial equilibrium model of firm cash flow by utilizing value function iteration. The model is specified with four parameters, α, δ, σ, and ρ, where σ represents the standard deviation of the productivity law of motion. The interest rate was set at 5 percent and the discount factor was calculated as 1/(1+r).

To estimate the model parameters, three different estimation methods were applied, including two simulation-based inference approach using SNPE and one approach using a SNRE GAN. The two SNPE approaches with the normalizing flow use slightly different embedding networks. These are the networks that convert the X' data into sufficient statistics to perform conditional estimation in the flow. The two methods are to use a feedforward neural network embedding or an recurrent neural network embeding (RNN), which has a inductive bias towards time series. Since the feed forward performed as well as the RNN and estimated faster, the RNN was dropped from future tasks. The results of the estimation are presented in the figures below

From the visual inspection of the charts, it can be observed that the posteriors obtained from the three estimation methods are similar, suggesting that the same distribution was learned. Additionally, the mode of the distribution appears to be close to the actual parameter values, which indicates that the model is capable of accurately recovering the true parameters.

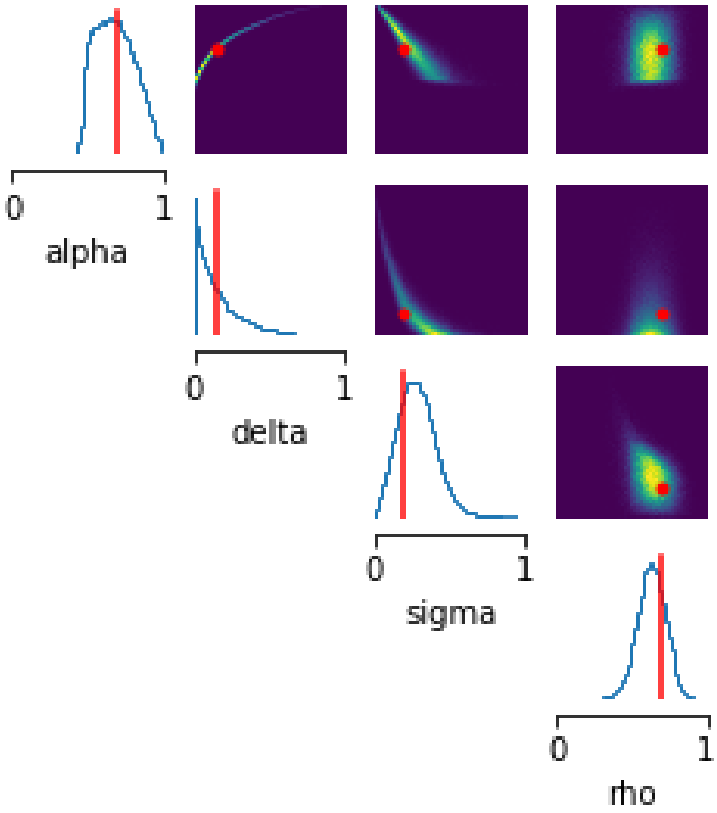

Figure V: Value Function Iteration with a Dense Net Embedding

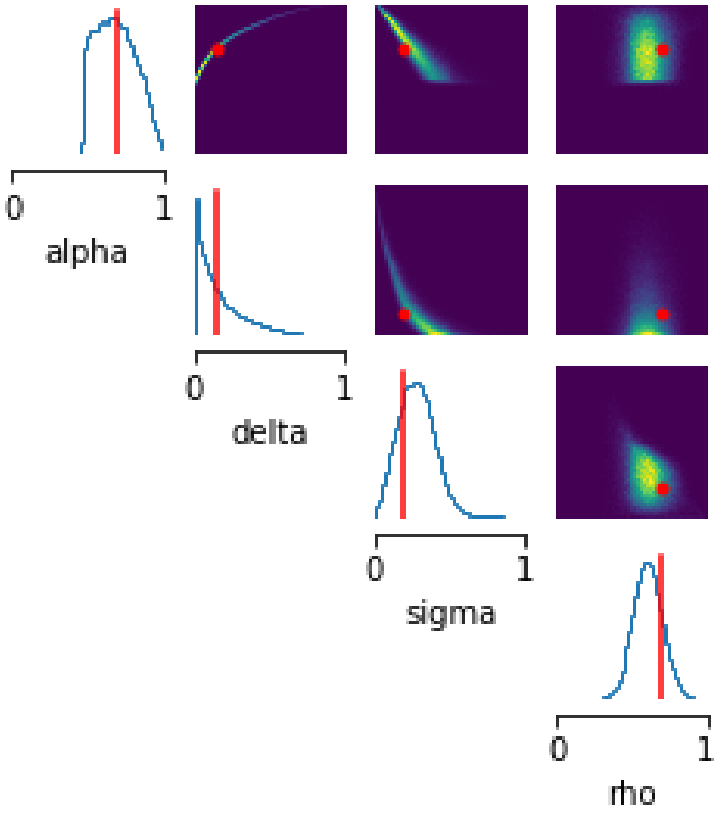

Figure VI: Value Function Iteration with an RNN Embedding

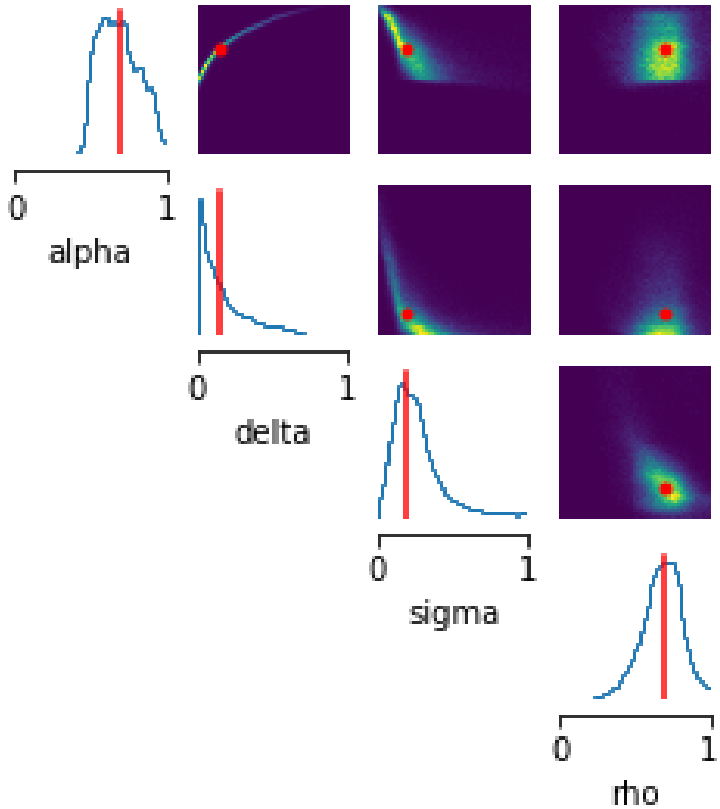

Figure VII: Value Function Iteration with a GAN Density Estimator

The use of the normalizing flow with dense and RNN embedding networks and the GAN density estimator highlights the versatility of this estimation framework and demonstrates the potential for future work in this area. The technical details of SNVI and its simplified version, Sequential Neural Ratio Estimation (SNRE), are discussed in the appendix for further reference.

IV.C. Lucas Asset Pricing Projection Model

In the following section, we demonstrate the estimation procedure on a Lucas asset pricing model, which is solved via projection without measurement error. The purpose of this estimation is to showcase the capability of the SBI approach to estimate models of this nature:

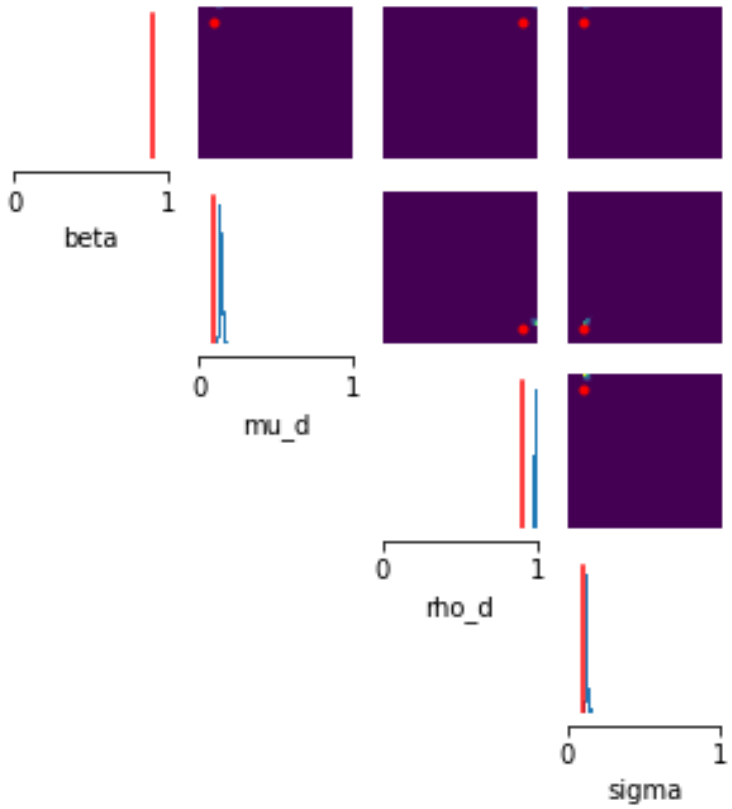

Figure VIII: Ten Parameter HANK model

The results of the estimation are presented in the above figure, which displays the ten-parameter HANK model. Upon visual inspection of the chart, it appears that the model accurately fits the true values, with the main error being observed in ρ_d, where the posterior is slightly higher than the actual value for the autoregressive parameters. All other posteriors match very closely with the actual data generating parameters.

This estimation procedure took under 10 hours to complete on an 8-core Intel i7 machine, highlighting the computational efficiency of this approach. This result also demonstrates that models solved via projection can be effectively estimated using the SBI framework.

IV.D. Smets-Wouters 2007 Model

The Smets-Wouters (2007) DSGE model is a representative agent New Keynesian model that describes an economy with a flexible price sector and a sector with sticky Calvo prices. The model has eight observed variables and eight shocks that interact with the rest of the model in an autoregressive manner. The linearized form of the model has 36 parameters, which were estimated using roughly 500,000 samples generated by the posterior. The parameters are separated into three different charts of 12 parameters each, and only the marginals are displayed to avoid visual clutter.

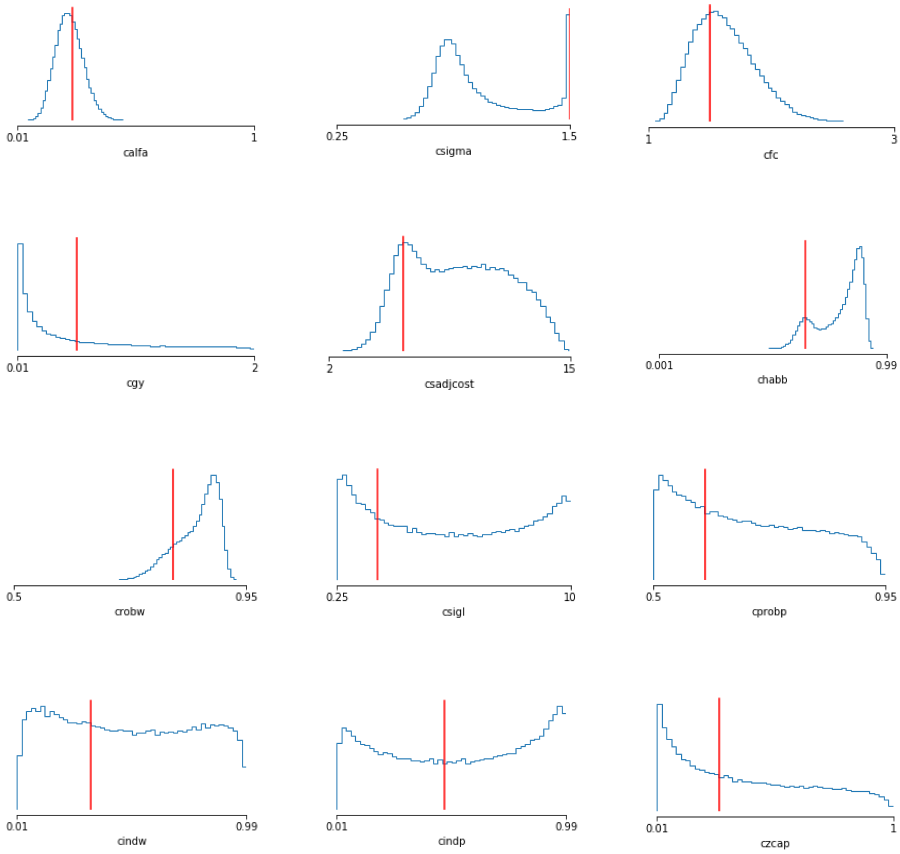

Figure IX: Smets-Wouters Posterior with Parameters 1-12

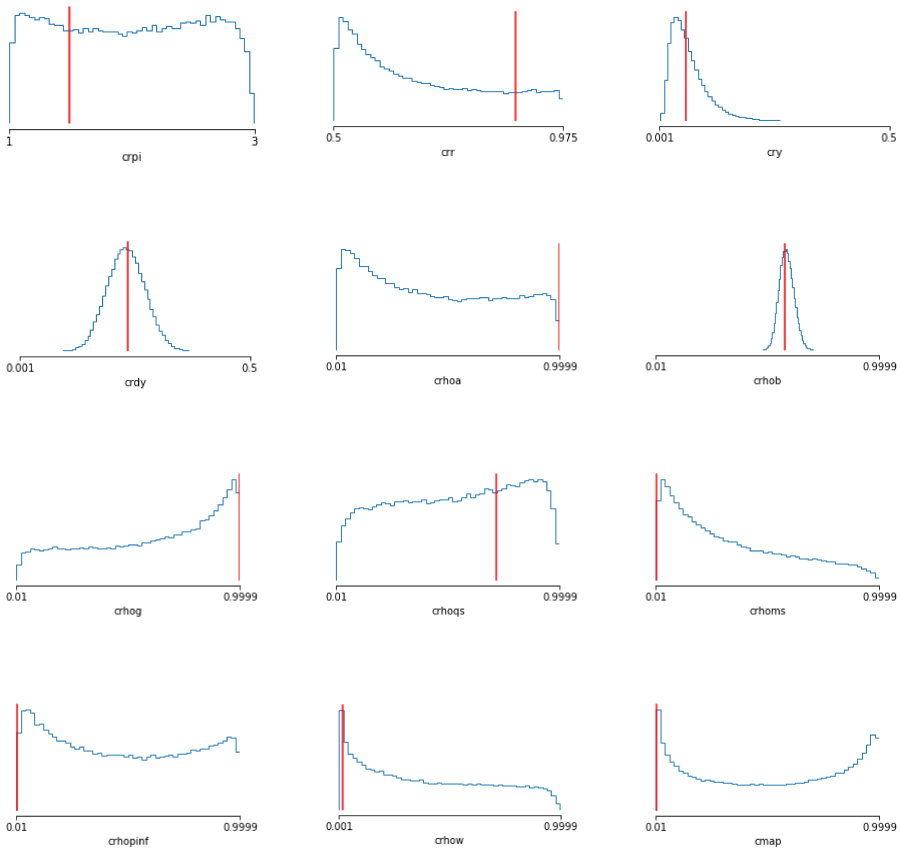

Figure X: Smets-Wouters Posterior with Parameters 13-24

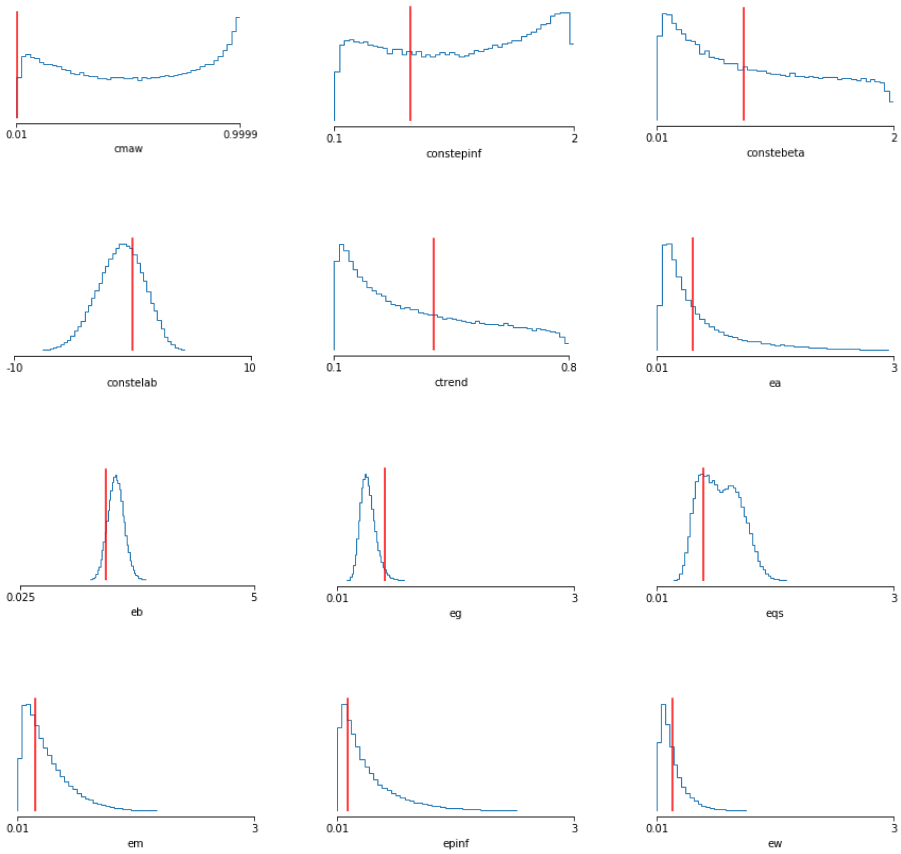

Figure XI: Smets-Wouters Posterior with Parameters 25-36

The Smets-Wouters model demonstrates varying levels of success in estimating the various parameters. For example, the model accurately estimates the capital share (calpha) but has more difficulty with the adjustments costs (csadjcost). The fixed costs (cfc) are estimated well. The Taylor rule parameters for output (cry) and change in output (crdy) are estimated well too, while the inflation Taylor rule parameter (crpi) is diffuse. Some of the autoregressive parameters for the shocks are well estimated, while others are diffuse or located in areas of low probability mass. The model performs well in estimating the standard deviation of the shock variables.

The posteriors estimated by the Smets-Wouters model with MH-MCMC are less successful in covering the true data generating process parameters. Despite the use of state-of-the-art Dynare

software (Adjemian et al., 2022) to draw samples, the model took roughly 30 hours to draw 10 million samples, and the posterior showed limited coverage even after ten thousand sample burn in per chain. Below I reproduce the (Smets and Wouters, 2007) using MH-MCMC instead.

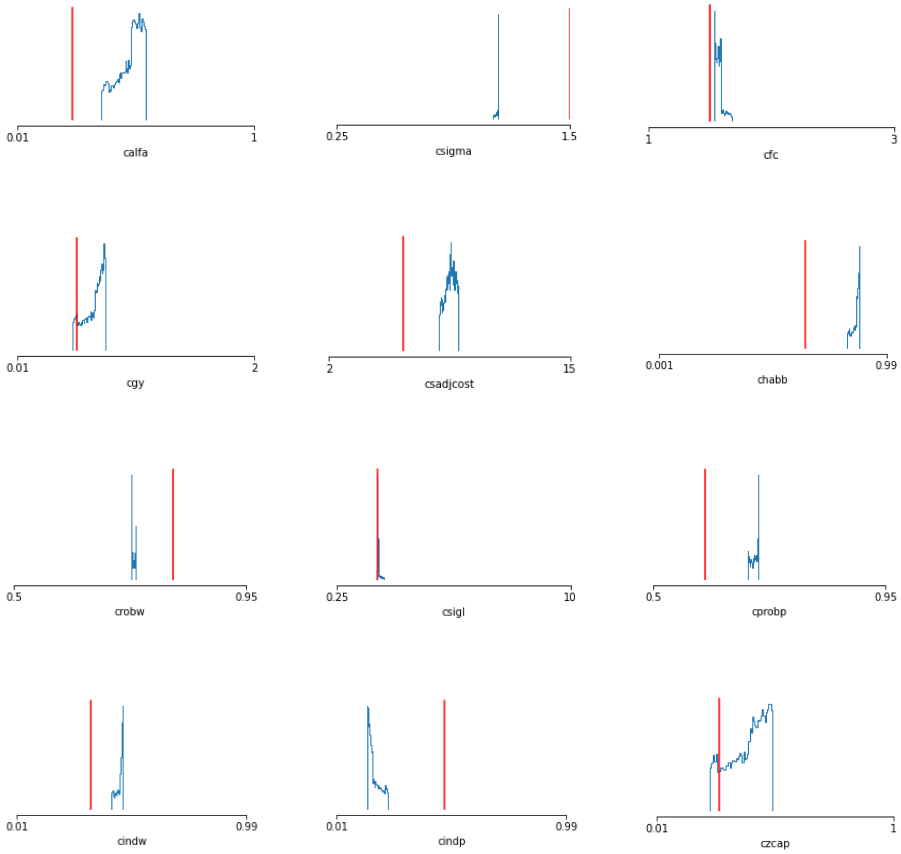

Figure XII: Smets-Wouters MH-MCMC Posterior with Parameters 1-12

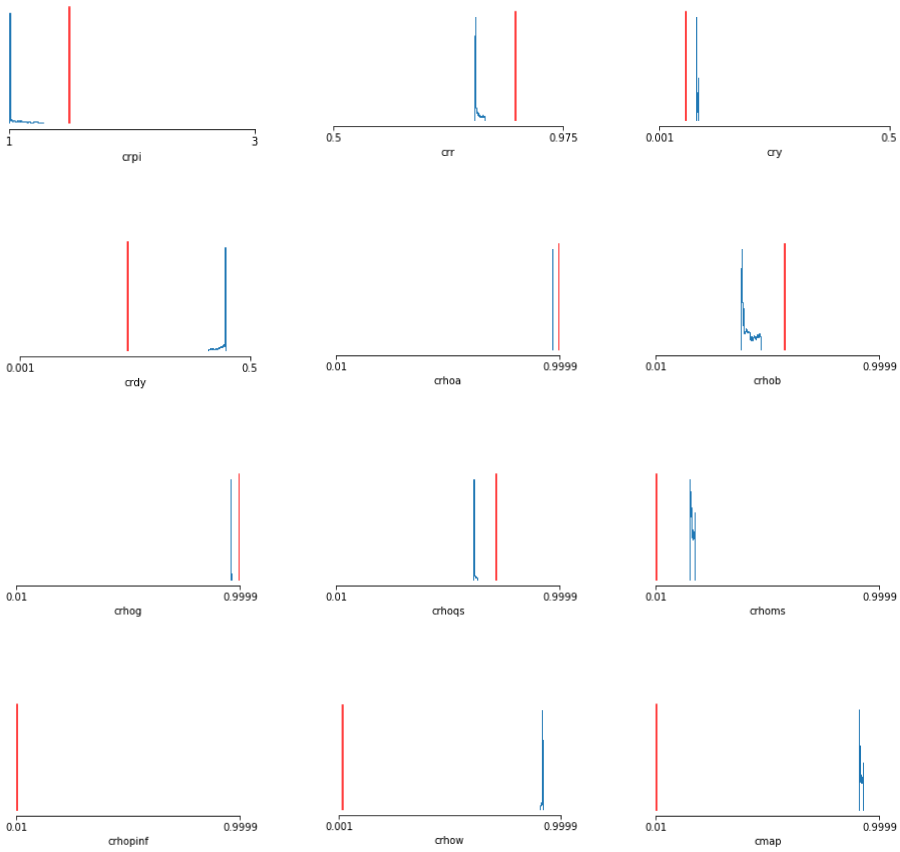

Figure XIII: Smets-Wouters MH-MCMC Posterior with Parameters 13-24

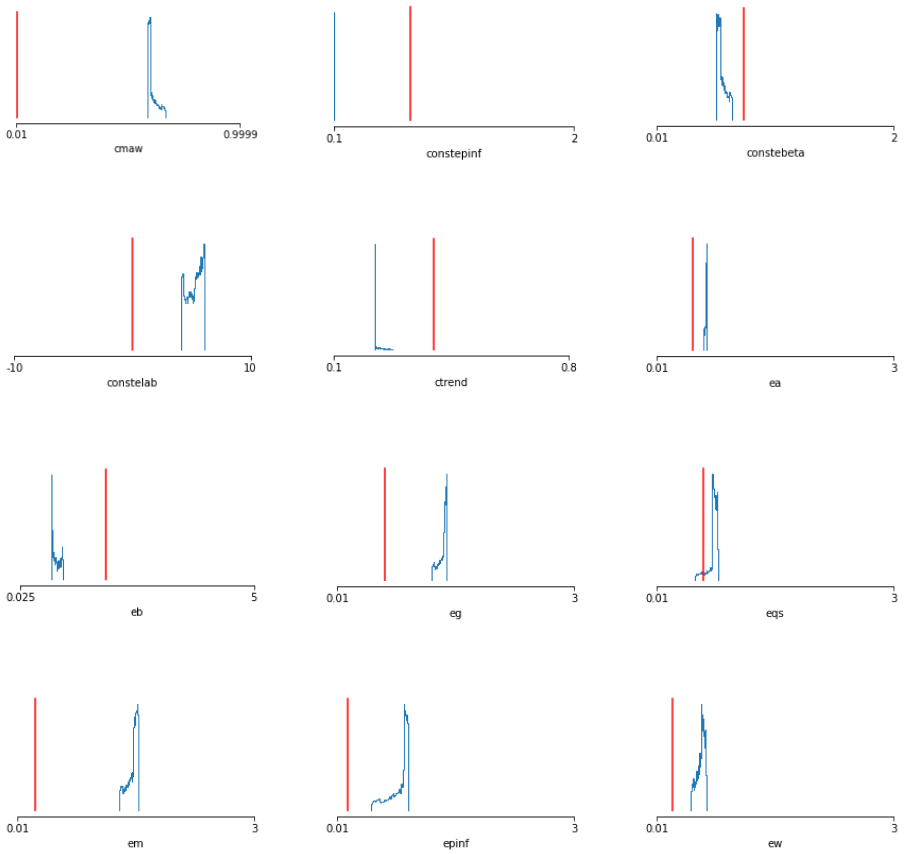

Figure XIV: Smets-Wouters MH-MCMC Posterior with Parameters 25-36

The performance of the MH-MCMC algorithm, as applied in this study, presents several challenges in terms of its ability to converge to the true data generating parameters. While some of the posteriors show good coverage of the true parameters, a substantial number of the posteriors exhibit a support outside of the ground truth and often converge to delta functions far from the true parameters. In particular, the crhow and cmap variables stand out as presenting the most glaring errors, with the MH-MCMC algorithm placing all its mass at around .99, whereas the true parameter is located at .015. On the other hand, the SBI posterior presents a more reasonable representation, putting some mass at .99 and also exhibiting a mode at .015 for both variables. This mistake is repeated to a lesser extent for other variables such as crdy, crpi, constepinf, con-

stelb, eb, em, csigma, chabb, cindp, among others. The posteriors in SBI have no issue reproducing multi-modal posteriors, a documented problem with MH-MCMC and is illustrated in the images.

The poor convergence of the MH-MCMC algorithm is believed, in part, to be due to the use of uninformative priors, especially those that are independent across marginal dimensions. Such priors structure the posterior in a way that makes it more independent across marginal dimensions, which, in turn, allows the marginally independent random walk kernel used in the MH-MCMC to better match the posterior. In contrast, uninformative priors do not influence the posterior to be more independent across marginal dimensions, resulting in strong covariance structures across axes, making the task of finding good proposal samples more challenging for the marginally independent random walk kernel. Even in the cases of informational priors, there is much evidence for the lack of convergence under realistic computational resources for MH-MCMC. For example, Herbst and Schorfheide (2014) demonstrate the identification problem for MH-MCMC in the Smets-Wouters model, while Andrews and Mikusheva (2015) highlight theoretical problems in identifying DSGE models in the MLE setting. When using MH-MCMC simulated annealing to perform MLE optimization, Siarry et al. (1997) demonstrate curse of dimensionalty issues even with objective functions with as few as 3-6 parameters

The SBI algorithm used in this study was implemented in Python, while the MH-MCMC algorithm was performed in Dynare with pre-loaded Smets-Wouters code from the authors, with only the priors changed to uniform priors and some estimation settings modified. However, due to the difference in programming languages and the overhead of calling Matlab code in Python, the estimation process was much slower for the SBI algorithm, with roughly 2 models solved per second, compared to about 100 models solved per second for the MH-MCMC. If the solution method in both algorithms were written in Python, MH-MCMC would be significantly slower than it is now. Tests indicate that MH-MCMC has a 3x to 5x advantage in terms of model solution speed Nevertheless, to ensure a fair comparison, the SBI posterior is shown below after 150 thousand samples, which took roughly the same 30 hours as the 10 million samples of the MH-MCMC:

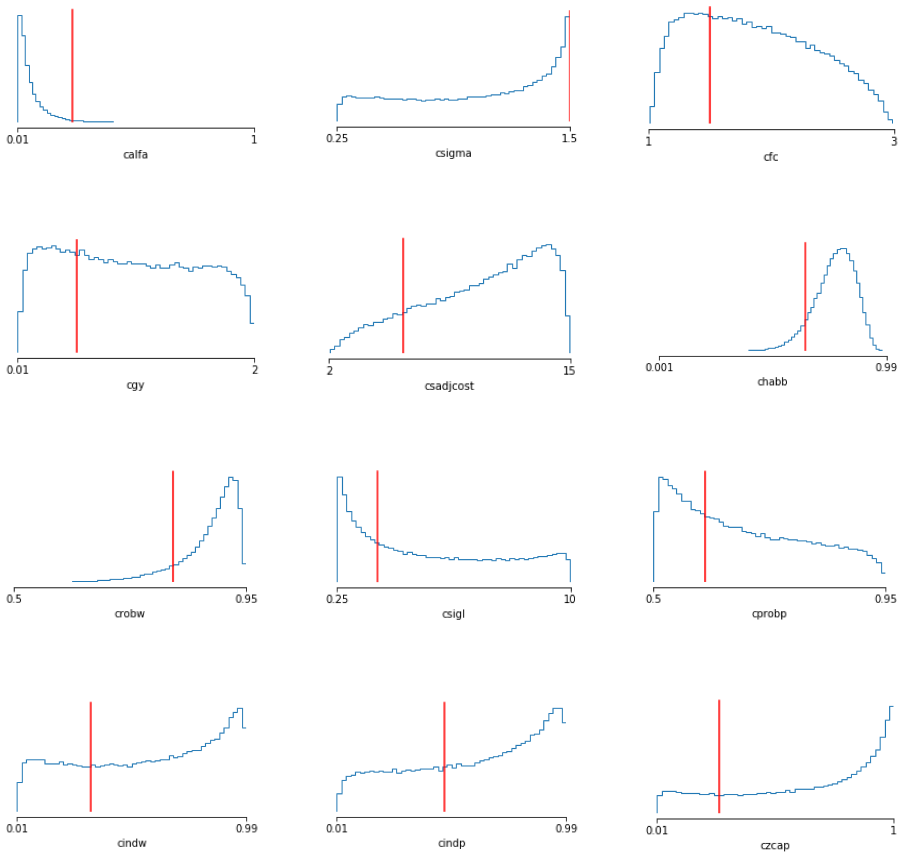

Figure XV: Smets-Wouters SBI Posterior with 150 Thousand Samples: Parameters 1-12

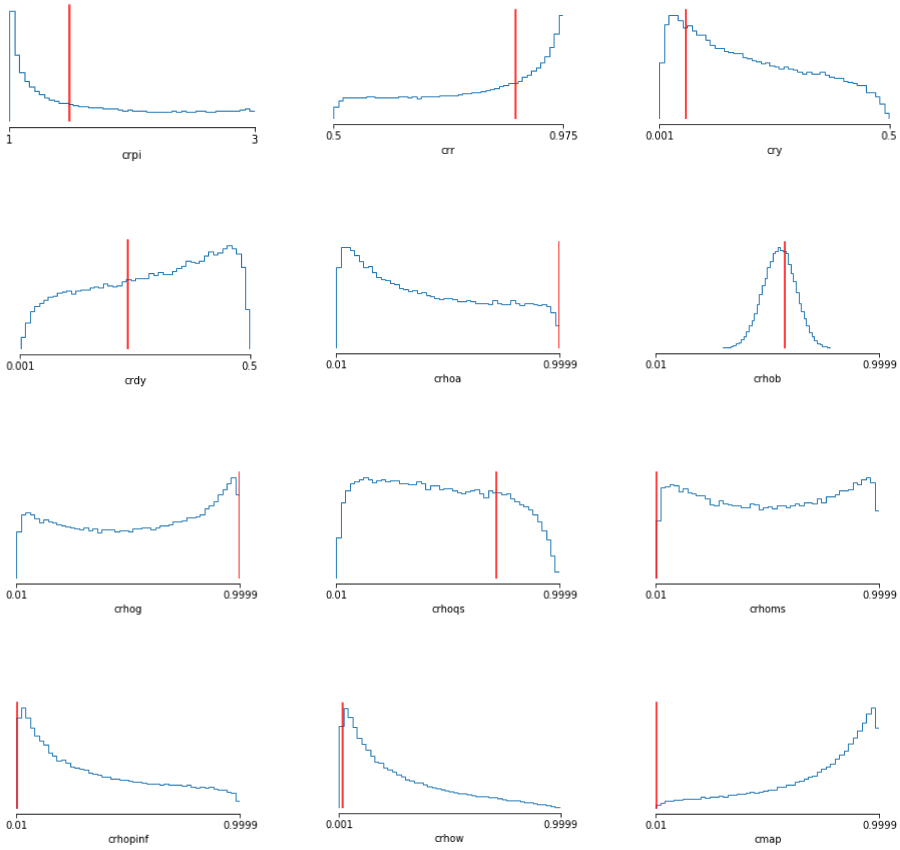

Figure XVI: Smets-Wouters SBI Posterior with 150 Thousand Samples: Parameters 13-24

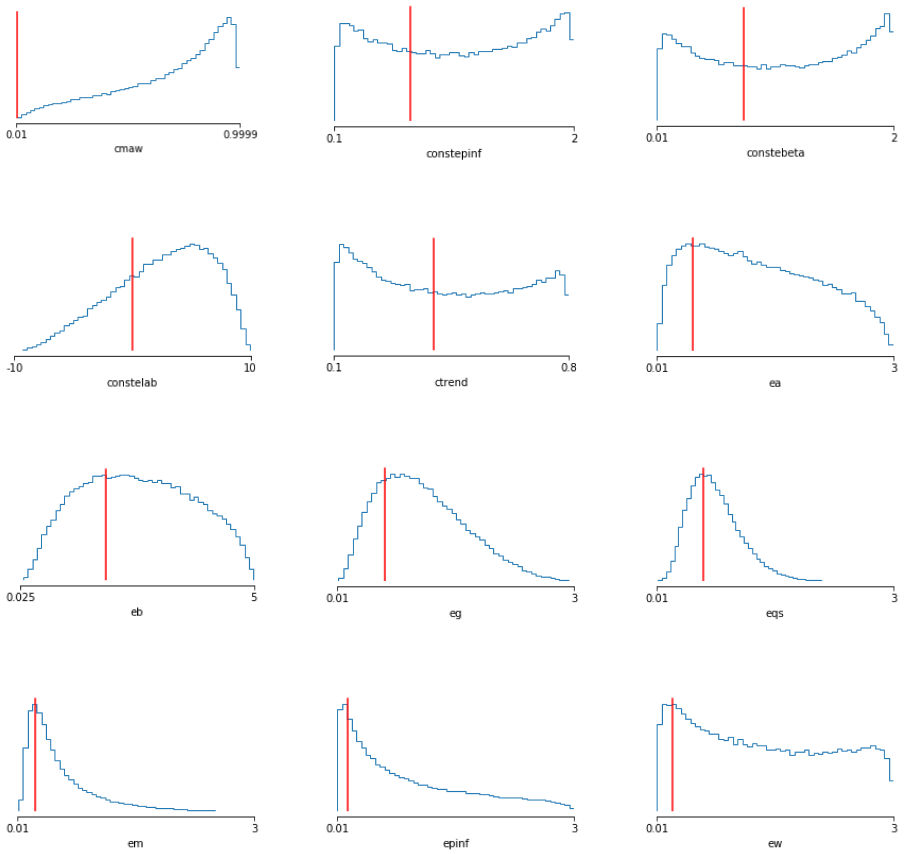

Figure XVII: Smets-Wouters SBI Posterior with 150 Thousand Samples: 25-36

Despite the slower sampling speed of the Python-based SBI algorithm, it still yields a more accurate posterior compared to the MH-MCMC algorithm. Although the SBI algorithm incurs a time penalty, it still manages to learn a better posterior in the same amount of time as the MH-MCMC. The performance of the SBI algorithm can be further improved by optimizing the Python code through techniques such as numba or writing the backend algorithm in C++, however, these are beyond the scope of this paper. The results indicate that SBI posterior is not perfect, however, it still assigns a significant probability mass to the true data-generating process and in some cases, even concentrates around the true parameters. This is a marked improvement compared to the MH-MCMC algorithm, which even after 10 million iterations, generally only produces highly

concentrated posteriors along a small range that often does not include the true data-generating parameter.

IV.E. HANK Model

The fifth model is the 10 parameter HANK model with the parameters discussed in the model section:

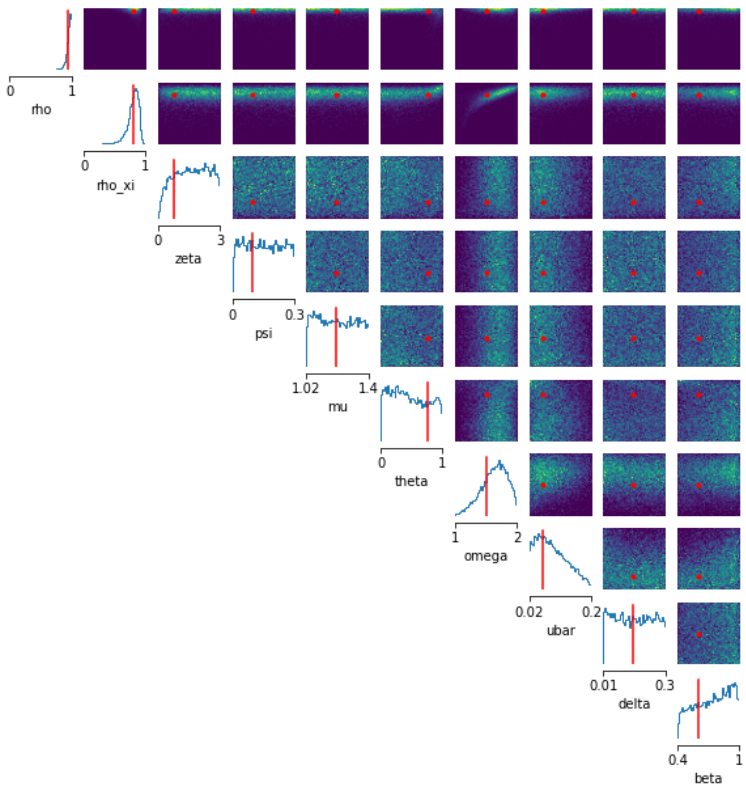

Figure XVIII: Ten Parameter HANK model–Partially Observed

The estimation of the HANK model is the focus of the next discussion. The HANK model features two observable shocks, namely the unemployment rate and the interest rate, and thus only two observable equations. Despite having a large latent state space of 810 latent variables, the estimation of the model was carried out successfully. The approach used in the estimation process was able to identify the parameters with confidence, and the estimated parameters were found to

be in agreement with the ground truth. This demonstrates the feasibility of estimating HANK models with a large latent state space, which is even computationally infeaible for an approach with a likelihood without using dimension reduction.

The approach used in the estimation process was able to estimate the parameters ρ, ρ_ξ, ω and $\bar{u}$ with high accuracy. The mode of the distribution of these parameters was found to be either at the true parameter value or in the case of ω, close to the true parameter value. However, for the other parameters, the likelihoods were found to be flat, which may be attributed to the limited impact of these parameters on the chosen observed variables. In conclusion, the estimation of the HANK model showed promising results, and highlights the potential of the approach used in the estimation process to estimate models with large latent state spaces.

IV.F. Winberry Heterogenous Agent Model

A sixth model is a Winberry (2018) approach to the Krusell and Smith (1998) model with full micromoments information as implemented in Liu and Plagborg-Møller (2023). My approach recovers similiar posterior results as Liu and Plagborg-Møller (2023), but my approach is around 13x faster per iteration as I only need micro and macro variable samples, without having the reconstruct the likelihood and use a sampling approach to numerically approximate an integral. I use the same number of iterations (10000) as Liu and Plagborg-Møller (2023), and get comparable results for my posterior. My approach even could estimate even more parameters and even use empirical data, however, the results suggest the model is unidentified in these settings, so I did not show my results. Liu and Plagborg-Møller (2023) do not show estimation results with more parameters and with empirical rather than simulated data.

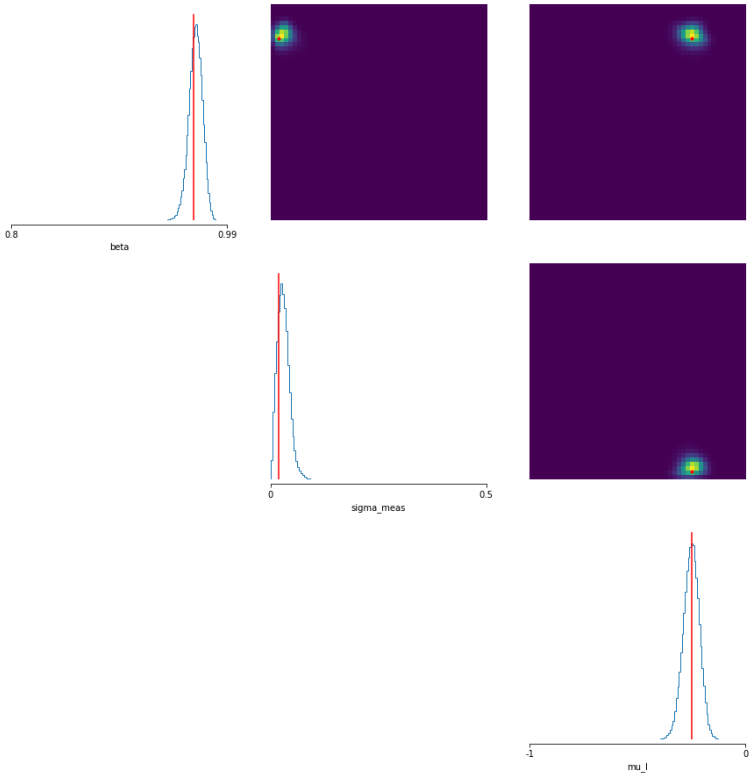

Figure XIX: SBI Results

Here are the Liu and Plagborg-Møller (2023) results:

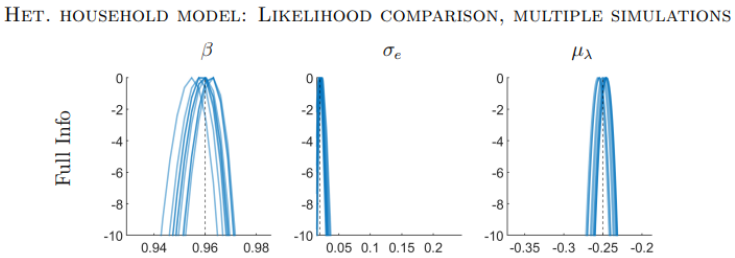

Figure XX: Liu and Plagborg-Moller Results

The results of the study are presented in the form of a figure above. The figure shows that the posterior concentrates almost entirely around the true parameter value, suggesting that the estimation was highly informative regarding the parameter value and accurately reproduces the

simulation parameters. The results are compared to those obtained by Liu and Plagborg-Møller (2023), which are presented in above as well. Both sets of results are similar and demonstrate the viability of the approach used in this study. This example illustrates how SBI is a fast and reliable approach to the estimation of heterogenous agent macroeconomic models.

IV.G. Bequests Model

The seventh model is the 11 parameter bequests model with parameters displayed in the chart corresponding to the parameters in the model appendix section:

The seventh model considered in this study is an 11-parameter bequests model. This chart corresponds to the parameters in the model appendix section of the paper.

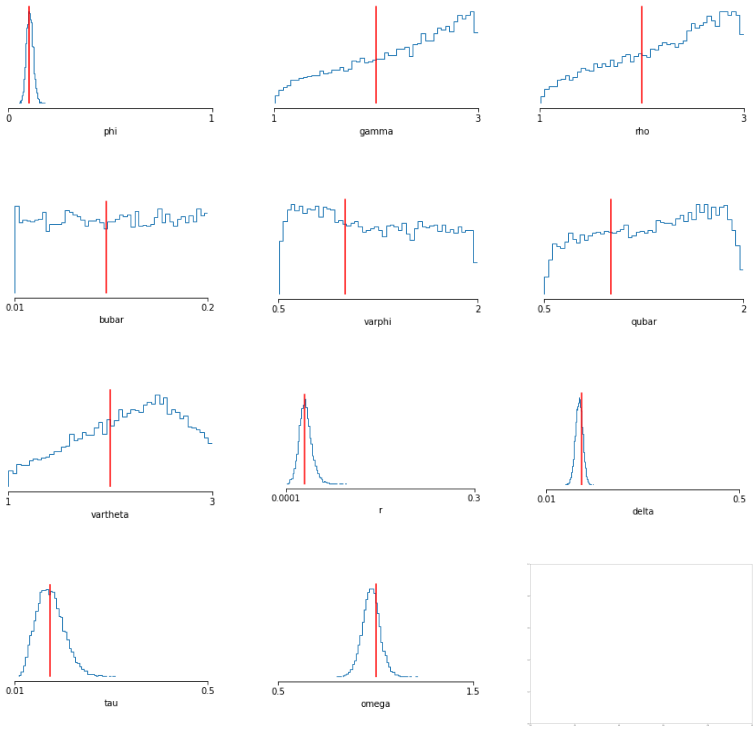

Figure XXI: Eleven Parameter Model Estimated via VFI

The results of the estimation of the model parameters via value function iteration (VFI) are shown in the figure. Although many of the parameters have converged to the true values, some of

the parameters still exhibit high uncertainty. This uncertainty could be due to the fact that the parameters have very flat likelihoods. Many of the posterior distributions do concentrate around the true values, suggesting that the model has some discerning power with regard to the parameters.

V. Conclusion

In this paper, I introduce a novel approach to estimate models in a Bayesian or MLE framework, which does not require a likelihood function. This approach enables the estimation of models while preserving the state space structure of the data generating process. The method has been applied to HANK models with large latent state spaces and has been successful in providing credible posterior estimates. Moreover, it can be applied to solution methods such as value function iteration and projection, which do not yield a likelihood function. This approach can be applied as broadly as MSM with improved guarantees for efficiency and posterior estimation.

The proposed method has a wider range of applications beyond macroeconomics. It can be used in the fields of industrial organization, applied game theory, and econometrics. In industrial organization and game theory, the method can be used to solve structural and dynamic models. In econometrics, it can be applied to estimate general latent time series models effectively and efficiently, without the need for a likelihood function. Additionally, the the suite of conditional density estimators can be used for density estimation in high dimensional settings.

In conclusion, the proposed method has a much broader scope of application than just macroeconomic modeling. The applications of this technique are not limited to this paper, and have the potential to be explored in greater depth in future research.

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A Economic Models

The models used to test these algorithms are disparate. I estimate a relatively standard RBC model via time iteration, a model of capital formation solved via value function iteration, a Lucas Tree model (Lucas Jr, 1978) solved with projection, the linearized Smets-Wouters 2007 model, a HANK model solved via Rieter's method with time iteration and lastly a 11 parameter bequests model solved via value function iteration. I will now discuss each of these models sequentially.

AA. The RBC Model

First I will estimate an RBC model. This RBC model is standard and comes from Alisdair McKay's note on heterogeneous agent models: `https://alisdairmckay.com/Notes/HetAgents/index.html`.

The utility function is a standard CRRA utility:

$$u(C_t) = E \sum_t \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \tag{1}$$

The production function is standard Cobb-Douglas with productivity:

$$Y_t = Z_t K_t^{\alpha} \bar{L}^{1-\alpha} \tag{2}$$

Where $\bar{L}$ is fixed labor and Z_t is productivity and K_t is capital which evolves according to a standard law of motion:

$$K_t = (1-\delta)K_{t-1} + Y_t - C_t \tag{3}$$

Productivity evolves according to a AR(1) process in logs:

$$logZ_t = \rho logZ_{t-1} + \epsilon_t \tag{4}$$

This model is solved via time iteration (Rendahl, 2017) and matches consumption, investment, productivity to real data, interest rates and capital are the latent unobserved variables. This is done on data both synthetic and real.

AB. The Model of Capital Formation

Next I will estimate a partial equilibrium model of capital formation. This model is a straightforward infinite horizon partial equilibrium model with risk neutral managers choosing investment to maximize the present value of a stream of firm cash flows and comes from solution code written by Emil Lakkis and Toni Whited.

Utility/cash flows are give by this formula:

$$U(K_t, I_t, Z_t) = \sum_t \beta^t * (\pi(K_t, Z_t) - I_t) \tag{5}$$

Where K_t, Z_t, and I_t are the same as in the RBC model–capital, productivity and investment respectively. π is the profit function.

The capital law of motion is like the RBC model:

$$K_t = (1 - \delta)K_{t-1} + I_t \tag{6}$$

Productivity evolves according to a AR(1) process in logs:

$$logZ_t = \rho logZ_{t-1} + \epsilon_t \tag{7}$$

Profit is defined as:

$$\pi(K_t, Z_t) = Z_t K_t^{\alpha} \tag{8}$$

This model shares similarities with the RBC model, with the main difference being the utility function and the value function solution method. This estimation method matches capital levels generated by a parameterized model.

AC. Lucas Asset Pricing Projection Model

' Next I will estimate a Lucas Tree model solved via projection. Following Mohammed Ait Lahren's code (`https://notes.quantecon.org/submission/5f44808ad24fdb001162a53b`), each agent maximizes CRRA utility:

$$max_i E_t[\beta^t \frac{c_t^{1-\gamma} - 1}{1-\gamma} \tag{9}$$

with γ set to 3. The law of motion is:

$$k_{t+1} = (1 + d_t/P_t)k_t - c_t/P_t \tag{10}$$

The state of the system is defined as dividends received plus price of the amount of assets owned:

$$m_{t+1} = (P_{t+1} + d_{t+1})k_{t+1} \tag{11}$$

Putting these three equations together one gets the pricing kernel:

$$p_t = \beta^t E_t[(\frac{c_{t+1}}{c_t})^{-\gamma}(d_{t+1} + p_{t+1})] \tag{12}$$

Dividends evolve under an $AR(1)$ process:

$$d_{t+1} = \mu_d + \rho_d * d_t + \epsilon \tag{13}$$

Where ϵ is a shock with standard deviation σ. When behaving optimally, consumption also equals the dividends received. This model matches the expected return and risk free rate at a variety of dividend levels, which demonstrates the ability of this model to not only estimate on models solved via projection, but also to work on cross sectional problems as well.

AD. Smets-Wouters 2007 Model

Additionally I will estimate a Smets Wouters 2007 model (Smets and Wouters, 2007). This section will mainly illustrate the equations and the parameters so one can cross reference parameters shown in the diagram with the correct equation in the DSGE model. This model comes from the paper Smets and Wouters (2007). More details on the economics of the equations can be found there. This model can be estimated both with MH-MCMC and SNPE. Thus this model mostly illustrates the increases in posterior accuracy when using an uninformative prior which goes hand in hand with estimation speed when using SNPE vs MH-MCMC.

First, a brief discussion of the overview of the model. The model is a representative New Keynesian model with stickiness in both wages and prices. The model has a flexible price set of equations which determines potential GDP for monetary policy and a sticky price system with a price markup equation that determines the true economy. I will only be discussing the sticky price system of equations with an implicit understanding that there is a mirroring flexible price system. Shocks all have autoregressive components to more accurately match data. All equations are linearized.

The aggregate resource constraint is:

$$y_t = ccy * c_t + ciy * i_t + \epsilon_t^g + 1 * crkky * z_t \tag{14}$$

The Euler equation is given by:

$$
\begin{aligned}
c_t =& (chabb/cgamma)/(1 + chabb/cgamma) * c_{t-1} + (1/(1 + chabb/cgamma)) * c_{t+1} + \\
& ((csigma - 1) * cwhlc/(csigma * (1 + chabb/cgamma))) * (l_t - l_{t+1}) - \\
& (1 - chabb/cgamma)/(csigma * (1 + chabb/cgamma)) * (r_t - \pi_{t+1}) + \epsilon_t^b
\end{aligned} \tag{15}
$$

The investment Euler equations are:

(16)

$$
i_t = (1/(1+cbetabar*cgamma))*(i_{t-1}+cbetabar*cgamma*i_{t+1}+(1/(cgamma^2*csadjcost))*q_t)+\epsilon_t^i
$$

The arbitrage equation for the value of the capital stock is:

$$
\begin{aligned}
q_t =& - r_t + \pi_{t+1} + (1/((1 - chabb/cgamma)/(csigma * (1 + chabb/cgamma)))) * \epsilon_t^b + \\
& (crk/(crk + (1 - ctou))) * r_{t+1}^k + ((1 - ctou)/(crk + (1 - ctou))) * q_{t+1}
\end{aligned} \tag{17}
$$

The linearized production function is:

$$
y_t = cfc * (calfa * k_t + (1 - calfa) * l_t + \epsilon_t^a) \tag{18}
$$

Capital utilization:

$$
k_t^s = k_{t-1} + z_t \tag{19}
$$

Capital utilization is some function of the interest rate for capital:

$$
z_t = (1/(czcap/(1 - czcap))) * r_t^k \tag{20}
$$

Capital's law of motion is:

$$
k_t = (1 - cikbar) * k_{t-1} + cikbar * i_t + cikbar * cgamma^2 * csadjcost * \epsilon_t^i \tag{21}
$$

The markup equation is:

$$\mu_t^p = calfa * k_t^s + (1 - calfa) * w_t - \epsilon_t^a \tag{22}$$

This equation is the only set of economic equations that doesn't have a flexible price counterpart. Note that the observed variable equations also will not have a flexible price counterpart.

The New Keynesian Phillips curve:

(23)

$$\begin{aligned}\pi_t =&(1/(1 + cbetabar * cgamma * cindp)) * (cbetabar * cgamma * \pi_{t+1} + cindp * \pi_{t-1}+\\ &((1 - cprobp) * (1 - cbetabar * cgamma * cprobp)/cprobp)/((cfc - 1) * curvp + 1) * (\mu_t^p))+\\ &\epsilon_t^p\end{aligned}$$

The rental rate of capital equation:

$$r_t^k = w_t + l_t - k_t \tag{24}$$

The path of wages is:

(25)

$$\begin{aligned}w_t =&(1/(1 + cbetabar * cgamma)) * w_{t-1} + (cbetabar * cgamma/(1 + cbetabar * cgamma)) * w_{t+1}+\\ &(cindw/(1 + cbetabar * cgamma)) * \pi_{t-1} - (1 + cbetabar * cgamma * cindw)/(1 + cbetabar * cgamma) * \pi+\\ &(cbetabar * cgamma)/(1 + cbetabar * cgamma) * \pi_{t+1}+\\ &(1 - cprobw) * (1 - cbetabar * cgamma * cprobw)/((1 + cbetabar * cgamma) * cprobw)*\\ &(1/((clandaw - 1) * curvw + 1)) * (csigl * l_t + (1/(1 - chabb/cgamma)) * c_t-\\ &((chabb/cgamma)/(1 - chabb/cgamma)) * c_{t-1} - w_t) + \epsilon_t^w\end{aligned}$$

The Taylor rule followed by central banks is:

$$r_t = crpi*(1-crr)*\pi_t+cry*(1-crr)*(y_t-y_t^p)+crdy*(y_t-y_t^p-y_{t-1}+y_{t-1}^p)+crr*r_{t-1}+\epsilon_t^r \tag{26}$$

The observed variables equations are:

$$dy_t = y_t - y_{t-1} + ctrend \tag{27}$$

$$dc_t = c_t - c_{t-1} + ctrend \tag{28}$$

$$di_t = i_t - i_{t-1} + ctrend \tag{29}$$

$$dw_t = w_t - w_{t-1} + ctrend \tag{30}$$

$$\pi_t^o = \pi_t + constepinf \tag{31}$$

$$r_t^o = r_t + conster \tag{32}$$

$$l_t^o = l_t + constelab \tag{33}$$

The autoregressive equations for shocks are:

$$\epsilon_t^a = crhoa * \epsilon_{t-1}^a + ea \tag{34}$$

$$\epsilon_t^b = crhob * \epsilon_{t-1}^b + eb \tag{35}$$

$$\epsilon_t^g = crhog * \epsilon_{t-1}^g + eg + cgy * ea \tag{36}$$

$$\epsilon_t^i = crhoqs * \epsilon_{t-1}^i + eqs \tag{37}$$

$$\epsilon_t^r = crhoms * \epsilon_{t-1}^r + em \tag{38}$$

$$\epsilon_t^p = crhopinf * \epsilon_{t-1}^p + epinfma - cmap * epinfma(-1) \tag{39}$$

$$epinfma = epinf \tag{40}$$

$$\epsilon_t^w = crhow * \epsilon_{t-1}^w + ewma - cmaw * ewma(-1) \tag{41}$$

$$ewma = ew \tag{42}$$

Some of the higher level parameters in the above equations are also related to the parameters in my diagram by the following equations:

$$cpie = 1 + constepinf/100 \tag{43}$$

$$cgamma = 1 + ctrend/100 \tag{44}$$

$$cbeta = 1/(1 + constebeta/100) \tag{45}$$

$$clandap = cfc \tag{46}$$

$$cbetabar = cbeta * cgamma^{(-csigma)} \tag{47}$$

$$cr = cpie/(cbeta * cgamma^{(-csigma)}) \tag{48}$$

$$crk = (cbeta^{(-1)}) * (cgamma^{csigma}) - (1 - ctou) \tag{49}$$

$$cw = \frac{(calfa^{calfa} * (1 - calfa)^{(1-calfa)}}{(clandap * crk^{calfa}))^{(1/(1-calfa))}} \tag{50}$$

$$cikbar = (1 - (1 - ctou)/cgamma) \tag{51}$$

$$cik = (1 - (1 - ctou)/cgamma) * cgamma \tag{52}$$

$$clk = ((1 - calfa)/calfa) * (crk/cw) \tag{53}$$

$$cky = cfc * (clk)^{(calfa-1)} \tag{54}$$

$$ciy = cik * cky \tag{55}$$

$$ccy = 1 - cg - cik * cky \tag{56}$$

$$crkky = crk * cky \tag{57}$$

$$cwhlc = (1/clandaw) * (1 - calfa)/calfa * crk * cky/ccy \tag{58}$$

$$cwly = 1 - crk * cky \tag{59}$$

$$conster = (cr - 1) * 100 \tag{60}$$

The seven observed equations dy_t, dc_t, di_t, dw_t, π_t^o, r_t^o, l_t^o, are matched using data simulated from a model with true parameters equal to the Smets-Wouters 2007 mean parameters, with slight modifications as some parameters were calibrated to 0.

AE. The HANK Model

Furthermore, I will estimate a HANK model solved via Reiter's method. This provides an example of estimation of models pushing the frontier in economics. The benefits of this approach allows one to estimate hetergenous agent models in a likelihood manner without resorting to demensionality reduction. The HANK model also comes from Alisdair McKay's tutorial, which adds a heterogeneous distribution of wealth as a state variable for households combined with a relatively standard New Keynesian model.

Preferences are still CRRA, like the RBC model. The firm side is now new Keynesian with intermediate goods produced via:

$$y_{j,t} = Z_t N_{j,t} \tag{61}$$

Here Z_t is aggregate productivity and N_t is labor employed by firm j producing variety $y_{j,t}$. The final good is produced via the Dixit-Stiglitz aggregator Dixit and Stiglitz (1977):

$$Y_t = (\int_0^1 {y_{j,t}}^{\frac{\epsilon-1}{\epsilon}})^{\frac{\epsilon}{\epsilon-1}} \tag{62}$$

Here ϵ is the elasticity of substitution between intermediates.

Output is defined as,

$Y_t = A_t \int_0^1 n_{j,t} dj = A_t N_t$

Where $n_{j,t}$ is the labor for the individual intermediate goods. Additionally from the accounting identity, output is also, $Y_t = C_t + \psi M_t H_t$ where the second term is hiring costs.

Government and bonds are defined by the following nominal and real budget constraint equations:

$$\tau P_t(w_t + d_t)(1 - u_t) + P_t B = (1 + i_{t-1}) P_{t-1} B + P_t b u_t \tag{63}$$

$$\tau(w_t + d_t)(1 - u_t) + B = R_t B + b u_t \tag{64}$$

Labor market is a standard McCall Search model:

$$N_t = (1-\delta)N_{t-1} + H_t \tag{65}$$

$$1 - u_t = (1-\delta)(1-u_{t-1}) + H_t \tag{66}$$

$$M_t = \frac{H_t}{u_{t-1} + \delta N_{t-1}} \tag{67}$$

$$M_t = \frac{1 - u_t - (1-\delta)(1-u_{t-1})}{u_{t-1} + \delta(1-u_{t-1})} \tag{68}$$

The household maximizes the CRRA utility like the RBC model. The households budget constraint in real money is:

$$a_t + C_t = R_t * a_{t-1} + earnings \tag{69}$$

Here a_t is savings and R_t is the interest rate as determined by the bond market clearing condition. Earnings are given by:

$$\text{earnings} = \begin{cases} (1-\tau_t)\,(w_t + d_t) & \text{if employed} \\ b & \text{if unemployed,} \end{cases} \tag{70}$$

Here w_t is wages, d_t is dividend income and b is unemployment benefits.

The stochastic matrix governing employment, unemployment transitions is given by:

$$\begin{pmatrix} 1 - M_t & \delta \\ M_t & 1-\delta \end{pmatrix} \tag{71}$$

Again M_t is governed by the labor market equations above.

The firm solves the Dixit-Stiglitz cost problem:

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon} Y_t \tag{72}$$

with price index P_t given by

$$P_t^{1-\varepsilon} = \int_0^1 p_{j,t}^{1-\varepsilon} dj. \tag{73}$$

The Calvo pricing adjustment parameter θ implies that the intermediate firm chooses price p^* to maximize:

$$\mathbb{E}_t \sum_{s=t}^{\infty} \theta^{s-t} R_{t,s}^{-1} \left[\frac{p_t^*}{P_s} y_{j,s} - w_s n_{j,s} - \psi M_s h_{j,s} \right] \tag{74}$$

Here $n_{j,t}$ denotes employment at firm j in time t. $h_{j,t}$ denotes hiring and $R_{t,s}^{-1}$ denotes the real interest rate discounted s periods back to time t. This model is solved using Reiter's method in combination with time iteration. This model matches unemployment and interest rate to the same data generated by a parameterized model, only two observed variables are used as the model only had two shocks.

AF. The Value Function Iteration Bequests Model

Finally, I estimate a model of bequests over time. The model illustrates the benefit of my approach as it cannot be estimated by perturbation because of the kink at the intersection of the consumption versus adjusting illiquid savings intersection. Thus one has to use simulation-based inference to Bayesian estimate this model solved via value function iteration.

This model comes from Jeppe Druedahl's open source model example: `https://github.com/pkofod/vfi/blob/master/Fast%20VFI.ipynb`. The utility function is:

$$u(b_t, c_t) = \frac{[\phi(b_t + \underline{b})^{1-\gamma} + (1-\phi)c_t^{1-\gamma}]^{\frac{1-\rho}{1-\gamma}}}{1-\rho} \tag{75}$$

Here b_t is bequest or non-liquid savings. c_t is consummation. b, ϕ, γ, ρ are all parameters. The bequest utility function (end of life utility) is:

$$\nu(a_t, b_t) = \varphi \frac{(a_t + b_t + \underline{q})^{1-\vartheta}}{1-\vartheta} \tag{76}$$

Here a_t is liquid savings and q, θ, φ are all parameters. The value function is give by:

$$v_t(m_t, n_t, l_t) = \max\{v_t^{keep}(m_t, n_t, l_t), v_t^{adj}(m_t, n_t, l_t)\} \tag{77}$$

$$\text{s.t.} \tag{78}$$

$$x_t = m_t + (1-\tau)n_t \tag{79}$$

Here m_t is beginning of period liquid savings (analogous to end of period a_t) and n_t is beginning of period illiquid savings (analogous to end of period b_t). x_t is the pooled value of total liquid and illiquid savings. l_t is labor income. The post-decision value function is:

$$w_t(a_t, b_t, l_t) = \nu(a_t, b_t), t = T \tag{80}$$

$$w_t(a_t, b_t, l_t) = \mathbb{E}_t\left[\max\{v_t^{keep}(m_{t+1}, n_{t+1}, l_{t+1}), v_t^{adj}(x_{t+1}, l_{t+1})\}\right], t < T \tag{81}$$

$$\text{s.t.} \tag{82}$$

$$l_{t+1} \sim F(l_t) \tag{83}$$

$$m_{t+1} = (1+r_a)a_t + \omega l_{t+1} \tag{84}$$

$$n_{t+1} = (1-\delta)b_t \tag{85}$$

$$x_{t+1} = m_{t+1} + (1-\tau)n_{t+1} \tag{86}$$

Here labor income comes from a distribution $F(l_t)$. Likewise r_a, ω, δ, τ are all parameters. The keep value function is:

$$v_t^{keep}(m_t, n_t, l_t) = \max_{c_t \in [0, m_t]} u(n_t, c_t) + \beta w_t(a_t, b_t, l_t) \tag{87}$$

$$\text{s.t.} \tag{88}$$

$$a_t = m_t - c_t \tag{89}$$

$$b_t = n_t \tag{90}$$

The keep value function is the value function at a point t if the agent chooses to consume and not

withdraw or deposit funds into the illiquid asset, b_t. The adjust value function is:

$$v_t^{adj}(x_t, l_t) = \max_{b_t \in [0, x_t]} v_t^{keep}(m_t, n_t, l_t) \tag{91}$$

$$\text{s.t.} \tag{92}$$

$$m_t = x_t - b_t \tag{93}$$

$$n_t = b_t \tag{94}$$

The adjust value function is the value of adjusting the illiquid asset. Notice the agent is unable to consume this period. This model with it's nonlinear and kinked decision rule cannot be effectively modeled with a method like perturbation. Even so, I use my estimation technique to get a Bayesian posterior distribution. This model matches the liquid asset, illiquid asset and consumption to synthetic data generated by a model.

B Background on Simulation Neural Ratio Estimators

BA. General Approach of Simulation-Based Inference

The differences between the two simulation-based inference approaches roughly centers around what conditional density estimator to use. Sequential Neural Ration Estimators (SNRE) uses a GAN and Sequential Neural Posterior Estimators (SNPE) use a normalizing flow. While one can sample a lot of points from the joint, an iterative approach where one fits the density estimator on sampled data then re-samples leads to lower variance. I will start discussing SNRE, by introducing the GAN

BB. Generative Adversarial Networks

While I focus on SNPE, there are related approaches that use a GAN (Goodfellow et al., 2014) as density estimators, Sequential Neural Ratio Estimation (SNRE). The GAN is an algorithm that uses two competing models moving in tandem to generate data that matches data generated from

the real world. Below is a diagram describing the structure of a basic GAN:

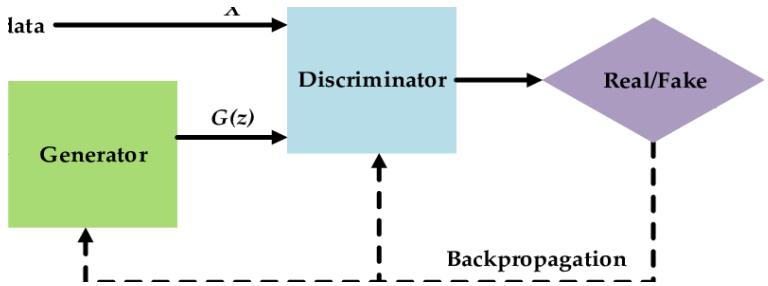

FIGURE XXII: DIAGRAM OF A GAN

The generator is a neural network (or any differentiable model like a dynamic stuctural model) which attempts to produce data (unconditionally or conditionally via some inputs Z) that resembles data in the real world. The discriminator is another neural network (or any trainable model that can produce a logistic output), which outputs the probability a given input data comes from the underlying data generating process and not from the generator. The two networks play a game where the objective of the generator is to convince the discriminator to assign high likelihood to its output and the objective of the discriminator is to assign the correct probabilistic label, a number between 0 or 1, to input that represents the continuum that the current data point is generated data or real data respectively. One way to think about a GAN is as a simulated method of moments algorithm where the moment conditions automatically evolve over many rounds to pick the most mismatched moments between the true data and the model. When thought in this way, it's intuitive to recognize that the GAN is nearly a full information and efficient estimation procedure. For more thorough proofs see Kaji, Manresa and Pouliot (2020).

Mathematically the discriminator loss is:

$$Loss_D = -E_{x \sim P_{data}(x)}[D(x)] - E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{95}$$

This is just a logit loss applied to data either coming from the data generating process or the generator. Likewise the generator's loss is to fool the discriminator:

$$Loss_G = E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{96}$$

These two loss functions can be compactly represented as:

$$G, D = \min_G \max_D E_{x \sim P_{data}(x)}[D(x)] + E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{97}$$

Equation (97) makes it clear the GAN objective is a minimax game with a solution that is a Nash equlibrium. However what is important for simulation-based estimation is to recover likelihoods/probabilities. The next proposition following Goodfellow et al. (2014) will show how to do that:

Proposition 1.: For a fixed generator, the optimal discriminator will produce a logistic output:

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_{generator}(x)} \tag{98}$$

Proof: Using equation (95), for the loss of the generator:

$$Loss_D = \int_x log(D(x))p_{data}(x)dx + \int_z log(1 - D(G(z)))p_{generator}(z)dx \tag{99}$$

$$= \int_x (log(D(x))p_{data}(x) + log(1 - D(x))p_{generator}(x))dx \tag{100}$$

(101)

Any discriminator that maximizes the integral: $\int_x (log(D(x))p_{data}(x) + log(1 - D(x))p_{generator}(x))dx$, maximizes the integral pointwise over all x. Thus the optimal discriminator will set the first order condition of the intergrand to zero:

$$0 = \frac{p_{data}(x)}{D^*(x)} + \frac{p_{generator}(x)}{1 - D^*(x)} \tag{102}$$

Rearranging equation (102) gets the result of the proposition. ■

Then one can use this proposition along with the discriminator output to estimate a likelihood ratio.

BC. Normalizing Flows

In this section, I will discuss how a particular deep learning model, the neural autoregressive flow (Huang et al., 2018), is structured. Then I will discuss how to use the change of variables formula for a random variable to derive a likelihood of a sample under the flow. I will also attempt to highlight advantages flows have over KDEs that make the SNPE algorithm more powerful that the traditional Kristensen and Shin (2012) approach, including the ability to do conditional density estimation, high dimensional density estimation, and the ability to sample and maximum likelihood estimate without resort to rejection sampling.

A normalizing flow, which the neural autoregressive flow is a member, is a composition of individual bijectors. Thus the first task is to define these intermediate changes in measure, $y^a, y^b ... y^z$, so that they can be composed with each other to form a invertable function. For example, the output of f_a becomes the input to f_b giving the total composition much more flexibility.

Invertability is not trivial. Take a polynomial regression. If one knows the input to this regression, one can calculate the output. However if one knows the output, the roots of a polynomial are not generally solvable and so polynomials aren't invertable transformations. However one set of invertable transformations are affine operations. Given $y^i, y^j \in R^N$ and n going from 1...N indexing each element in the vector y^i:

$$\forall n, y^j_n = \beta^0_n + \beta^1_n * y^i_n \tag{103}$$

This is trivially invertable as one can derive y^i from y^j by shifting the negative of β^0 and scaling by the inverse of β^1. While one can still add nonlinearities like a logit link function between bijective transformations, this still lacks expressivity. The standard way to extend this model is to allow for the shifts and scales to be dependent on at least some of the inputs. For example, earlier elements $y^i_{a:l}$ are only modified in an directly invertable manner (ie $y^i_{a:l} = \beta^0 + \beta^1 * y^j_{a:l}$), then condition the shifts and scales for the rest of the elements $y^i_{l:N}$ on inputs, $y^i_{a:l}$, that weren't transformed:

$$\forall \iota \in 2 : N, y^j_\iota = \beta^0_\iota(y^i_{a:\iota-1}) + \beta^1_\iota(y^i_{a:\iota-1}) * y^i_\iota \tag{104}$$

$\beta^0(.)$ and $\beta^1(.)$ are typically differentiable and flexible functions–typically feed-forward neural networks conditioned on earlier elements of the vector, that now output dynamic "psuedo" parameters β^0, β^1 that depend on the conditioning variables, $y^i_{a:\iota-1}$.

Next I will illustrate how a neural autoregressive bijector is defined that I use in my paper. Make the first element, y^j_1 the identity or affine map, then the second element y^j_2 an affine transformation in y^i_2 with psuedo-parameters $\beta_0(y^i_1)$ and $\beta_1(y^i_1)$ dependent only on y^i_1. The third element, y^j_3, has psuedo-parameters then dependent only on $y^i_{1:2}$ and y^i_3 enters only in an affine manner.

$$y^j_\iota = \sigma(\beta^0_\iota(y^i_{a:\iota-1}) + \beta^1_\iota(y^i_{a:\iota-1}) * y^i_\iota)$$

To make this model nonlinear, one typically adds a link function σ which is usually a Leaky ReLU (Xu et al., 2015). A Leaky ReLU is two lines of different slope intersecting at zero. This kink is often enough to make neural networks universal approximators of continuous functions (Cybenko, 1989).

Since σ is invertable as well as the rest of the flow, one can recover the inputs y^i knowing only all elements in y^j. The procedure to recover y^i_2 from all the y_j's is to know y^i_1 from the affine first equation. Then recover the affine parameters for the y^i_2 equation (which is only conditioned on y^i_1). Then one can invert y^j_2 to get y^i_2 knowing the affine parameters. Now that one knows $y^i_{1:2}$ one can repeat the procedure to get y^i_3, etc. This makes the bijector invertable in practice. This invertability is another advantage normalizing flows have over KDE. One can sample from them

and calculate the pdf of any sample without the use of rejection sampling or other computationally heavy techniques. This will be exploited in the SNPE algorithm as the flow will be used both as a density estimator and as a proposal distribution for generating θ samples.

Furthermore, if one stacks multiple bijections on top of one another, one can permute the order of input elements ($y^i_{1:N}$) with a bijective operation so that different bijectors have different conditioning relationships among variables allowing the flow to universally approximate any distribution. If the functions for $\beta^0(.)$ and $\beta^1(.)$ are universal approximators (ie neural networks), one can show that a stack of bijectors (along with permutations), is also a universal approximator and thus can approximate any change of variable arbitrarily well (Huang et al., 2018).

One can also condition $\beta^0(.)$ and $\beta^1(.)$ with additional arbitrary conditioning variables to form a flexible conditional density estimator. This ability to condition is one advantage it has over a KDE. Since the problem is to estimate the density of θ conditioned on x, θ will take the role of $\{y^a_1..y^a_n\}$. X will be the true data and analogue of simulated data which will be differentiated by a lower case x. Conditioning variable x and real data X will enter as additional conditioning variables in the flow. In particular they will be included as conditioning variables in the neural networks that produces $\beta^0(.)$ and $\beta^1(.)$. Thus, for example β^1_5 will be conditioned on $y^i_1...y^i_4$ as well as x/X, the the accompanying simulated or real data.

Next I will discuss the change of variable formula for a flow. Given a density, $p(y^a)$ with random vector realizations y^a; a normalizing flow is a function, $f(y^a)$ mapping y^a to a target random variable y^z, which has density, $q(y^z)$ such that the probability $q(y^z)$ is related to the probability $p(y^a)$ via:

$$q(y^z) = p(y^a)|det\frac{df^{-1}}{dy^a})| = p(y^z)|det\frac{df(y^a)}{dy^a}|^{-1} \tag{105}$$

This formula is simply the change of variable formula one learns in an introductory PhD econometrics class, with the first term $p(y^a)$ representing the measure in the base distribution, and the Jacobian representing the change in measure due to the transformation to the target distribution $q(y^z)$. For example, the quadratic function is a change of variable for a normal to a chi-squared distribution, and in order to derive the measure of a chi-squared distribution, one needs to not

only transform the realizations of the normal distribution via the quardatic function, but also take into account the Jacobian of this transformation. Huang et al. (2018) and Rezende and Mohamed (2015) show how a neural autoregressive flow is both a bijection and can approximate any continuous distribution. Assuming both these things, one can use the change of variable formula to derive the likelihood that a sample point y^z was generated by the flow (Rezende and Mohamed, 2015). One uses the flow to transform the y^z to y^a and then one can use $p(y^a)$ and the change of variable Jacobian to calculate $q(y^z)$. One can then perform maximum likelihood by modifying the parameters of the neural networks that govern the psuedo-parameters ($\beta_i^0(.)$ and $\beta_i^1(.)$ in each layer to maximize $q(y^z)$ for y^z samples in the data. Since flows can model any conditional distribution (Huang et al., 2018), and can calculate the likelihood of any sample point in the target space, one can fit this model on samples from arbitrary continuous densities and have the guarantees that come with a well specified maximum likelihood problem. Thus, normalizing flows are density estimators that are robust to misspecification and asymptotically efficient due to estimation with maximum likelihood. Additionally, since each affine transformation is a function of variables that have an index smaller than the index in question, the Jacobian is lower triangular and the determinant is just the product on the diagonal. This makes the change in variable formula scale linearly with the number of parameters.

BD. Sequential Neural Ratio Estimation

In SNRE, Durkan, Murray and Papamakarios (2020) use a GAN, not to match a underlying data like in proposition 1, but to contrast between two different distributions to estimate a density. In macro, the dataset is typically a single data point with correlated timesteps. Using a GAN to discriminate between real data and fake data would not work because the GAN would learn a delta function for the real data. In order to ameliorate this, the authors use contrastive learning where they differentiate between the generated distribution and the independent distribution, $x, \theta \sim p(x)p(\theta)$, where x is sampled uniformly from the empirical distribution of simulated data. Then they contrast the independent distribution, with the true joint: $x, \theta \sim p(x, \theta) = p(x|\theta)p(\theta)$. Furthermore, knowing the optimal discriminator gives the ratio discussed in (98). The authors then recover the

ratio: $r(x,\theta) = \dfrac{p(x,\theta)}{p(x)p(\theta)}$, using the fact that $\dfrac{p(x,\theta)}{p(x,\theta)+p(x)p(\theta)}$ equals $\dfrac{r}{r+1}$. Thus, the density of the posterior $p(\theta|x) = r(x,\theta)p(\theta)$, where $p(\theta)$ is the prior. This extends Kaji, Manresa and Pouliot (2020) GAN approach in allowing estimation of models with a latent time dimension. One can extend this to multi-round inference where one uses the GAN as a better proposal distribution than the prior, by sampling θ from the GAN using MCMC. Then one uses importance sampling to deal with the mismatch in proposal distribution and prior. See Durkan, Murray and Papamakarios (2020), Hermans, Begy and Louppe (2020) and the SNRE theoretical sections of the appendix for more information.

Algorithm 2: SNRE Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, discriminator $d_\phi(x,\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim p^{(i-1)}$ for $n = 1...S$ with MCMC;

  Simulate $x^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{x^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(x,\theta)$ *not converged* **do**

    Sample $\{x^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Sample K contrasting samples for each i sample $\{x'^{(k)}\}_k^{B*K}, \{\theta'^{(k)}\}_k^{B*K} \sim D\{x\}, D\{\theta\}$, breaking the joint distribution of D into independent marginals;

    Train the GAN using the GAN loss function from the two distributions x, θ and x', θ' and a softmax/multinomial logit objective $\sum_i^B log(\dfrac{exp(d_\phi(x^{(i)}, \theta^{(i)})}{\sum_k^K exp(d_\phi(x^{(k)}, \theta^{(k)})})$;

  end

  Update posterior $p^{(i)} \propto exp(d_\phi(x,\theta))p(\theta)$;

end

BE. Sequential Neural Variational Inference (SNVI)

In SNVI, Glöckler, Deistler and Macke (n.d.), one performs SNRE, but instead of MH-MCMC sampling from the GAN posterior, one projects the distrbiution derived by the GAN unto a normalizing flow with variational inference. As discussed in this article, MH-MCMC takes a long time to converge to the posterior. Since one can calculate the pdfs of a sample point using the GAN ratio estimation, one can sample from a normalizing flow, calculate the pdf of the sample under both the flow and the GAN, and use this to minimize the KL divergence between the flow and the GAN. Thus the flow inherits the distribution of the GAN, and one can sample directly from the flow without resorting to MH-MCMC

Algorithm 3: SNVI Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, discriminator $d_\phi(x,\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim f_{\phi'}^{(i-1)}(\theta)$ for $n = 1...S$;

  Simulate $x^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{x^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(x,\theta)$ *not converged* **do**

    Sample $\{x^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Sample K contrasting samples for each i sample

    $\{x'^{(k)}\}_k^{B*K}, \{\theta'^{(k)}\}_k^{B*K} \sim D\{x\}, D\{\theta\}$, breaking the joint distribution of D into independent marginals;

    Train the GAN using the GAN loss function from the two distributions x, θ and x', θ' and a softmax/multinomial logit objective $\sum_i^B log(\frac{exp(d_\phi(x^{(i)}, \theta^{(i)})}{\sum_k^K exp(d_\phi(x^{(k)}, \theta^{(k)})})$;

  end

  Train normalizing flow with variational inference

  $f_{\phi'}^i = argmin_{\phi'}(KL(f_{\phi'}^i(\theta)||exp(d_\phi(x,\theta))p(\theta))$;

end

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